



Introduction to Heat Transfer and Steady State Heat Conduction

Syllabus :

Introduction : Heat transfer mechanism, Conduction heat transfer, Thermal Conductivity, Convection heat transfer, Radiation heat transfer, Laws of heat transfer

Steady state conduction : General heat conduction equation, Boundary and initial conditions, One-dimensional steady state conduction : the slab, the cylinder, the sphere, composite systems .

Section I : Introduction and Basic Concepts

Thermodynamics and Heat Transfer with Application Areas of Heat Transfer

1.1 Thermodynamics and Heat Transfer

In day to day life, we are encountered with transfer of various forms of energy and their conversion from one form to another. We have defined **energy** as its capacity to do work i.e. its capacity to produce an effect especially to change of motion. The conversion of energy from one form to another may be partial or complete. It is based on the laws of thermodynamics.

The **science of thermodynamics** deals with the energy transfer and its effect on the condition of the system. Though the energy exists in various forms, however in engineering applications we are more concerned with the conversion of heat energy into work and vice-versa. However, in this text, we are primarily concerned with transfer of heat energy.

The **heat** is defined as the form of energy which can be transferred from one system to another system across their boundaries due to temperature difference existing between the two systems. The amount of heat energy transferred across the system can be determined by the application of **first law of thermodynamics** involving work and other forms of energies.

During heat transfer, it is observed that the heat energy always flows in direction from higher temperature medium to lower temperature medium and the transfer of heat energy stops once both the mediums reach to their equality of temperature. Thus the **driving force of transfer of heat energy is temperature difference and the rate of heat**

transfer increases with the increase in temperature gradient/difference.

However, there are many engineering applications where it is essential to know the rates of heat transfer and the temperature distribution under steady and transient conditions in designing the various components of a system. *The science which deals with the determination of rates of transfer of heat energy is known as **heat transfer**.*

From the above discussions, it is evident that there exists a major difference between thermodynamics and heat transfer. The thermodynamics deals with the relation between the heat energy and the other forms of energy especially the work and internal energy. While the heat transfer is primarily concerned with the rates of heat transfer only.

The knowledge of the subject of heat transfer helps us to determine the following :

- The rates of heat transfer at given temperature difference.
- Time taken to affect the desired amount of heat energy.
- The surface area needed to transfer given amount of heat energy and
- The temperature distribution across the system.

This knowledge gained ultimately leads us to the design of various components/systems.

1.1.1 Applications of Heat Transfer

The knowledge of the heat transfer is applicable to all branches of engineering. Some of the engineering applications where the design of systems are based on the rates of heat transfer are :

- Design of heat exchangers, ducts etc in refrigeration and air conditioning.



2. Design of cylinders, radiators etc in internal combustion engines.
3. Design of combustion chambers and cooling of blades for gas turbine power plants.
4. Design of motors, generators, transformers etc.
5. Design of steam generators, condensers, cooling towers and other heat exchangers in thermal and nuclear power plants.
6. Design of solar collectors and other related equipments.
7. Design of furnaces, space vehicles and various components of chemical processes etc.
8. Various electronic systems like computers, TV, VCR etc.
9. To prevent creation of thermal stresses in concrete structures during curing.
10. Heat treatment of metals to obtain the desired properties of materials etc.

1.2 Modes of Heat Transfer

The heat can be transferred from one substance to another substance by the following three modes :

1. Conduction ; 2. Convection
3. Radiation

All modes of heat transfer require the existence of temperature difference and the heat is always transferred in the direction of decreasing temperature.

Conduction of heat can be defined as the transfer of heat from one part of the substance to another part of the same substance without appreciable motion of molecules. The conduction can take place in solids, liquids and gases.

The process of heat **convection** is due to the capacity of moving matter to carry heat energy. The transfer of heat by convection takes place between a solid surface and the adjacent liquid or gas layer that is in motion. In case the fluid is at rest then the transfer of heat between the solid surface and the adjacent layer of fluid is purely by conduction.

The process of heat transfer by **radiation** is due to the electromagnetic radiation emitted in a wavelength band between 0.1μ to 100μ ($1 \mu = 10^{-6} \text{ m}$) solely as a result of the temperature of a surface. Therefore, the transfer of heat energy of radiation does not require the presence of any material medium as in case of heat transfer by conduction or convection.

In a physical problem, rate of heat transfer is controlled by the combined effect of all the modes of heat transfer i.e. by conduction, convection and radiation in varying degree. However, solution of a problem on heat transfer involves the identification of the major modes of heat transfer as compared to other modes of negligible heat transfer.

To illustrate how all the three modes are involved, let us consider the case of transfer of heat from the furnace of boiler to the water flowing in the tubes. In this case the heat is dissipated by the flue gases to metal surface of tube both by radiation and convection. This heat is further transferred by conduction across the thickness of tube to its inner surface and further it is transferred by convection and radiation to water in the tubes.

1.3 Mechanism of Heat Transfer by Conduction

The process of heat conduction has been defined as the transfer of heat energy through the substances without any appreciable motion of the molecules from the region of high temperature to the region of lower temperature.

This mode of heat transfer by conduction is accomplished via the following two mechanisms :

- (i) By Lattice Vibrations
- (ii) By Transport of free Electrons

(i) By Lattice Vibrations

- The molecules of a substance continuously vibrate in random orientation about same mean position. These vibrations are called as *lattice vibrations*.
- We know that the kinetic energy (K.E.) of the molecules in case of liquids and gases is due to their random translational, rotational and vibrational motions. However, the solids only vibrate in their lattice. The temperature of the substance corresponds to its K.E. i.e. higher is the average K.E. of molecules, higher will be the temperature of the substance.
- The molecules of solid materials while vibrating, they collide with each other and the molecules having higher K.E., transfer some its energy by impact to adjacent molecules having lower K.E. This type of energy transfer will continuously takes place through substance as long as there exists a temperature gradient. Therefore,

- "The rate of heat transfer due to lattice vibration depends upon the rate of collision between the molecules."

(ii) By Transport of free Electrons

- The mechanism of heat conduction and the mechanism of transport of electric current are both highly dependent upon the flow of free electrons.
- The valence electrons in the outer most orbit of an atom get excited on availability of energy. They overcome the binding force to become free and move within the boundaries of the solid. These electrons are called *free electrons*. A large number of free electrons move about in their lattice and impart their energy from higher level to lower level.
- The process of heat conduction occurs in all substance viz solids, liquids and gases. *The pure conduction is found only in solids*. The solids which are good electric conductors have large number of free electrons which move and vibrate in their lattice. These solids are called **metals**. *For this reason the materials which are good electric conductors are also good heat conductors e.g. silver, copper, aluminium etc.*

1.4 Basic Definitions Relating to Heat Transfer

1. Heat Transfer Rate and Heat Flux
2. Steady State
3. Unidirectional Heat Flow

1.4.1 Heat Transfer Rate and Heat Flux

- **Heat** has been defined as the form of energy in transit, for which the driving force is the temperature difference. Heat is never contained in a body, it is only recognized when it crosses the boundaries of a system.
- The amount of heat energy transferred during a process over a given period of time is denoted by letter Q (Joules) while *the amount of heat transferred per unit time is called heat transfer rate, Q (J/s)*. Therefore,

$$Q = \int_0^t Q \cdot dt \quad \text{Joules} \quad \dots(1.4.1)$$

where, 't' represents the time interval during which the total heat transfer Q is to be determined.

- The rate of heat transfer per unit area normal to the direction of heat transfer is called the **heat flux rate, q** i.e.

$$q = \frac{Q}{A} \quad (\text{W/m}^2) \quad \dots(1.4.2)$$

where 'A' represents the heat transfer area.

Note : 'Q' has been used in general in the text as heat transfer rate until it is specified.

1.4.2 Steady State

- A system is said to be under **steady state** if the temperature within the system at any specific location or surfaces remains constant and it does not change with time.
- Since the internal energy of a system is the function of temperature, it implies that the internal energy of the system remains constant under steady state. It also suggests that the rate of heat energy into the system must be equal to the rate of heat energy out of the system. For example the energy transfer in heat exchangers, flow of current in conductors, flow of fluids in pipes etc.
- Under **unsteady state** heat transfer, the temperature varies with time, thus it is the function of both location and time. For example the cooling of automobile engine, and freezing of food in refrigerators, heat treatment of metals by quenching etc.

1.4.3 Unidirectional Heat Flow

- A system said to have **unidirectional heat flow** if the heat transfer in other directions is negligible e.g. transfer of heat in a slab or plate whose temperature depends on time and varies only in one direction.
- Other examples are the heat flow in pipes and spheres in which the temperature varies in radial direction.
- Thus we can define the unidirectional heat flow as the flow of energy along any one of the axes with respect to the co-ordinate system used for evaluation of heat transfer rates.

1.5 Fourier's Law of Heat Conduction

Fourier in 1822 had proposed the empirical law of heat conduction based on general observations which is called as the **Fourier's law of heat conduction**.

**Statement**

It can be stated as follows :

"The rate of heat flow through a simple homogeneous solid is directly proportional to the area measured normal to the direction of heat flow and the temperature gradient in the direction of heat flow".

For the heat flow in X-direction, mathematically it can be expressed as :

$$Q \propto A \cdot \frac{dT}{dx} \quad \dots(1.5.1)$$

where,

Q = Heat transfer rate (J/s or W) in given direction.

A = Area of heat flow normal to heat flow direction (m^2).

dT = Temperature difference between two faces of a block of thickness dx ($^{\circ}C$ or K).

dx = Thickness of solid body (m).

$\therefore \frac{dT}{dx}$ = Temperature gradient in the direction of heat flow (K/m).

Thus, $Q = -k \cdot A \cdot \frac{dT}{dx} \quad \dots(1.5.2)$

Heat flux, $q = \frac{Q}{A} = -k \cdot \frac{dT}{dx} \quad \dots(1.5.3)$

where k is the sign of proportionality called **coefficient of thermal conductivity of material**.

Negative sign in Equation (1.5.2) appears due to the fact that dT is negative in the direction of heat flow i.e. +ve X-direction because the heat always flows from higher temperature to lower temperature. It also implies that the value of k is positive (Refer Fig. 1.5.1).

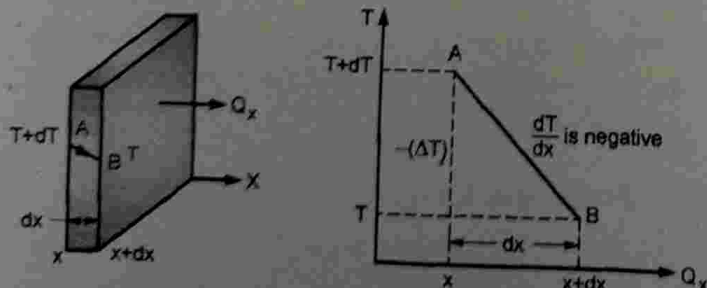


Fig. 1.5.1

1.5.1 Units of Thermal Conductivity, k

Thermal conductivity k has units which can be determined from Equation (1.5.2) as follows :

$$k = \frac{Q \cdot (W) \times dx (m)}{A (m^2) \times dT ({}^{\circ}C \text{ or } K)} \\ = \frac{W}{m K} \text{ or } \frac{W}{m {}^{\circ}C} \quad \dots(1.5.4)$$

1.5.2 Definition of Thermal Conductivity of a Material

Thermal conductivity is the ability of material to conduct heat through it. It can also be defined as the amount of heat flow rate per unit area normal to the direction of heat flow through unit thickness of the material per unit temperature difference.

1.5.3 Thermal Conductivity of Materials

Thermal conductivity varies widely for various engineering materials and it is the function of temperature, density, structure etc.

Thermal conductivity of few materials is given in Table 1.5.1.

Table 1.5.1 : Thermal conductivity of materials, k at $0^{\circ}C$

Sr. No.	Material	Thermal conductivity k (W/mK)
1.	Metals	
	Silver (pure)	417
	Copper (pure)	386
	Aluminium (pure)	202
	Magnesium	171
	Tungston	166
	Zinc (pure)	112
	Brass	108
	Iron (pure)	73
	Tin	65.9
	Wrought-iron (0.5% C)	59
	Steel (C = 0.5%)	55
	Steel (C = 1%)	43
	Steel (C = 1.5%)	36
	Lead	35.1

Sr. No.	Material	Thermal conductivity k (W/mK)
2.	Non-Metallic solids	
	Marble	2.1 to 2.9
	Brick common	1.34
	Brick, fire clay	0.96
	Concrete	0.88
	Glass	0.75
3.	Liquids	
	Mercury	8.20
	Water	0.556
4.	Ammonia	0.540
	Engine oil (SAE50)	0.147
	Freon-12 (CCl ₂ F ₂)	0.073
	Gases	
	Hydrogen	0.169
	Helium	0.141
	Water vapour (Saturated)	0.0205
	Carbon di-oxide	0.0146
	Air	0.024
	Insulations	
5.	Asbestos	0.046
	Glass fibre	0.043
	Cork	0.039
	Felt blanket	0.03
	Polystyrene	0.027
	Urethane foam (RUF)	0.026
	Silica powder	0.0017

1.5.4 Important Points Regarding Thermal Conductivity of Materials

Following important points may be noted with regard to thermal conductivity of various materials under different conditions :

1. Thermal conductivity in materials
2. Order of thermal conductivity in various materials
3. Mechanism of heat conduction
4. Effect of Variation of Temperature on Thermal Conductivity

5. Effect of temperature on thermal conductivity of gases
6. Effect of Temperature on Thermal conductivity of liquids
7. Anisotropic and Isotropic Materials
8. Dependence of Thermal conductivity k on Temperature
9. Assumptions in Fourier's law of heat conduction

1. Thermal conductivity in materials

Thermal conductivity of metals is mainly due to flow of free electrons while in case of other solids and fluids it is due to molecular vibration/collisions. (Refer section 1.3).

2. Order of conductivity in various materials

Thermal conductivity of materials in decreasing order is as follows (Refer Table 1.5.1) :

Metals → Non-Metallic solids → Liquids → Gases

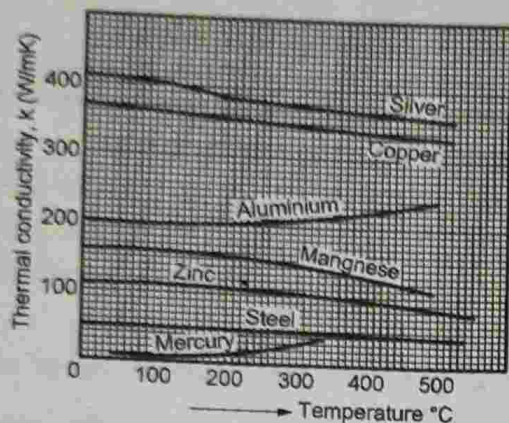
3. Mechanism of heat conduction

- The heat transfer in solids is both by transport of free electrons and by lattice vibrations.
- But the transfer of thermal energy in solids (metals) mainly depends upon the transport of free-electrons, for this reason, *the materials which are good electrical conductors are also good heat conductors.*

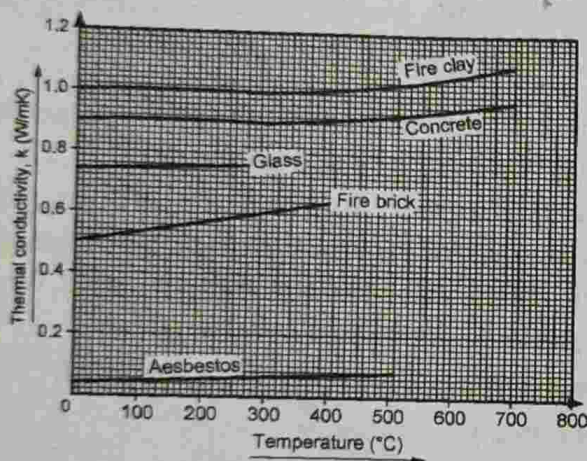
4. Effect of Variation of Temperature on Thermal Conductivity

- The heat conduction in solids is both by transport of free electrons and lattice vibrations. When the temperature of metals increases, the lattice vibrations impede the motion of free electrons.
- Due to this the thermal conductivity of most of the pure metals decreases with increase in temperature.
- Most non-metals are poor conductors of heat transfer thus they have low thermal conductivity and are called as **thermal insulators.**
- **Whereas the thermal conductivity of alloys and insulating materials,** having few free electrons, increases with increase in temperature because their heat conduction largely depends upon the lattice vibrations. It implies that $(k)_{\text{metal}} \propto \frac{1}{T}$ and $(k)_{\text{non-metal}} \propto T$.

- Note that mercury is an exception for the above statements. The thermal conductivity of mercury increases with the increase in temperature.
- The variation of thermal conductivity of metals and non-metals with change in temperature is shown in Fig. 1.5.2(a) and (b) respectively.



(a) Metals



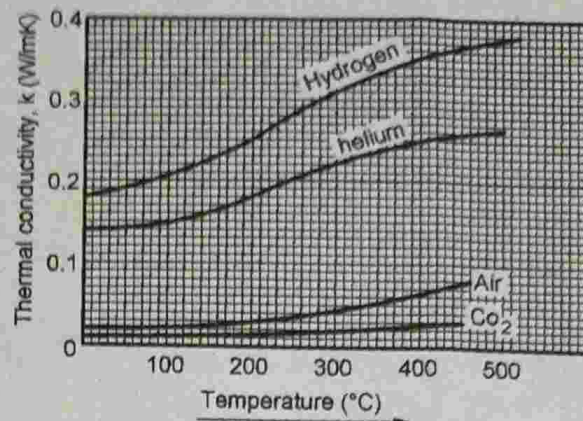
(b) Non-metals

Fig. 1.5.2 : Variation of thermal conductivity of solids with temperature

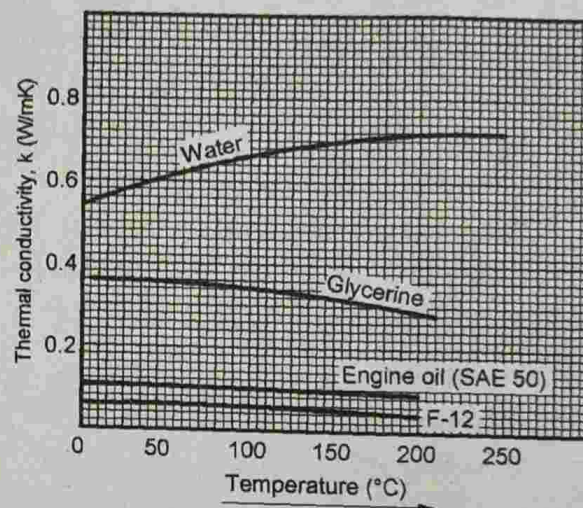
5. Effect of temperature on thermal conductivity of gases

- The transport of heat energy by conduction in liquids and gases is due to random motion of molecules exchanging energy and momentum.
- The kinetic energy of molecules is the function of temperature. When the molecules of higher temperature region collide with molecules of lower temperature region, they lose their K.E by collisions.
- Therefore in case of gases the thermal conductivity of ideal gases increases with the increase in temperature since at higher temperatures the molecules will have higher rate of collisions.

- The variations of thermal conductivity of gases with change in temperature is shown in Fig. 1.5.3(a).



(a) Gases



(b) Liquids

Fig. 1.5.3 : Variation of thermal conductivity of gases and liquids with variation in temperature

6. Effect of Temperature on Thermal conductivity of liquids

- The mechanism of heat conduction in liquids is complex but it is similar to gases. However, the molecules of a liquid are more closely spaced compared to gases and the molecular force field exerts a strong influence on energy exchange between molecules. Thus the thermal conductivity of liquids is both the function of pressure and temperature.
- It is observed that the thermal conductivity of liquids tends to decrease with increase in temperature (water being an exception) due to decrease in density with increase in temperature. The variation of thermal conductivity of liquids with change in temperature is shown in Fig. 1.5.3(b).

7. Anisotropic and Isotropic Materials

- Thermal conductivity also depends on grain structure of materials. Some materials like wood have different thermal conductivity in X, Y, Z directions. Such materials are called **an-isotropic** materials.
- The materials having same thermal conductivity in all directions are called **isotropic** materials.

8. Dependence of Thermal conductivity k on Temperature can be written as

$$k = k_0 (1 + \alpha \cdot T) \quad \dots(1.5.5)$$

where k_0 = Thermal conductivity at 0°C (W/mK)

α = temperature coefficient of thermal conductivity (per $^\circ\text{C}$)

T = temperature ($^\circ\text{C}$)

' α ' is positive for non metals and insulating materials and ' α ' is negative for pure metals.

9. Assumptions in Fourier's law of heat conduction are

- Heat flow is unidirectional under steady state conditions.
- The temperature gradient is linear and constant.
- There is no internal heat generation.
- The material is homogeneous and isotropic.

1.6 Heat Conduction Through a Wall/Slab

Consider a wall of surface area A of thickness x as shown in Fig. 1.6.1. Let Q be the heat transfer rate in X-direction as shown and ' k ' be the thermal conductivity of material.

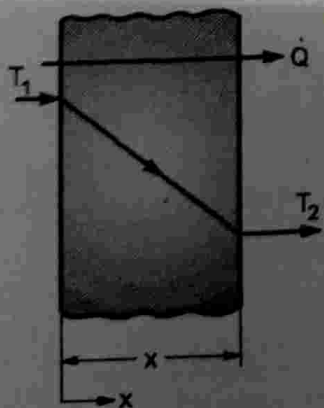


Fig. 1.6.1 : Heat conduction through a wall

Rewriting the Fourier's law of heat conduction from Equation (1.5.2),

$$\text{Heat transfer rate, } Q = -k \cdot A \cdot \frac{dT}{dx}$$

Integrating between boundary conditions :

$$(i) \text{ At } x = 0, T = T_1$$

$$(ii) \text{ At } x = x, T = T_2$$

$$Q \int_0^x dx = -k \cdot A \int_{T_1}^{T_2} dT$$

$$Q \cdot x = -k \cdot A (T_2 - T_1)$$

$$\text{or, } Q = k \cdot A \frac{(T_1 - T_2)}{x} \quad \dots(1.6.1)$$

$$\text{Heat flux, } q = \frac{Q}{A} = k \frac{(T_1 - T_2)}{x} \quad \dots(1.6.2)$$

1.7 Convection

It is observed that the fluids are always in direct physical contact with solids. Due to this, heat transfer always takes place from the heated solid surface to the colder fluid or from the hotter fluid to the cold solid surface as a consequence of the motion of the fluid relative to solid surface.

The process of heat transfer between the solid surface and a fluid flowing past the surface is called **convection**.

In order to understand the concept of convective heat transfer, let us consider a hot vertical wall in contact with cold air as shown in Fig. 1.7.1.

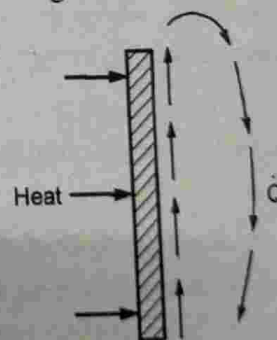


Fig. 1.7.1 : Natural or free convection

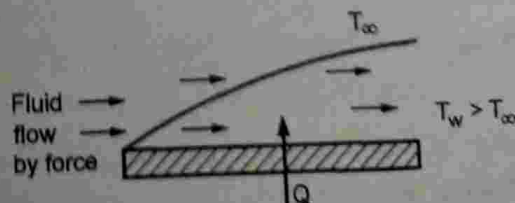


Fig. 1.7.2 : Forced convection

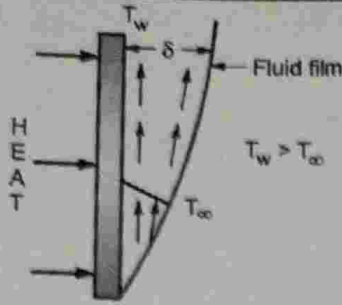


Fig. 1.7.3 : Heat conduction through a fluid film

It will be observed that with the lapse of time, the layer of cold air in contact with surface is heated by heat conduction through the wall surface and the density of air decreases. Due to density difference, a buoyant force results and it causes the heated air to rise which is replaced by the fresh cold air. This process is repeated continuously and sets up natural convection current. This method of heat transfer is called the **natural or free convection** which is due to the circulation of fluid by virtue of natural difference in densities of cold and hot fluids.

To improve upon the rate of convective heat transfer, the external force such as fan and blowers are used to set up convection currents as shown in Fig. 1.7.2. In this case the heat transfer is called by **Forced convection**.

1.7.1 Coefficient of Convective Heat Transfer

In case of heat transfer by convection (natural or forced), there exists a very small fluid film adjacent to the wall surface of thickness δ , where the temperature varies from wall surface temperature, T_w to outer film temperature, T_∞ as shown in Fig. 1.7.3.

Heat is transferred by conduction through film from wall surface to cold air.

Let k_f be the thermal conductivity of this film. Then we define the **coefficient of convective heat transfer or film conductance**, h as the ratio of thermal conductivity of the film to its thickness. Therefore, coefficient of convective heat transfer,

$$h = \frac{\text{Thermal conductivity of film, } k_f}{\text{Film thickness, } \delta} \quad \dots(1.7.1)$$

The value of δ decreases with increase in fluid velocity. For this reason the convective heat transfer by forced convection is more than the heat transfer by natural convection.

The value of k_f is low for gases as compared to the liquids.

1.7.2 Newton's Law of Cooling

The rate of convective heat transfer between a surface and the fluid was first suggested by Newton and is known as **Newton's law of cooling**.

The Newton's law of cooling states that the rate of heat transfer is proportional to the surface area perpendicular to heat flow direction and the temperature difference between the wall surface temperature T_w and the fluid temperature, T_∞ in the direction perpendicular to heat flow direction.

$$Q \propto A \cdot (T_w - T_\infty) \quad (\text{Assume } T_w > T_\infty)$$

$$\text{or } Q = h \cdot A \cdot (T_w - T_\infty) \quad \dots(1.7.2)$$

Where 'h' is the constant of proportionality called the coefficient of convective heat transfer or film conductance.

Units of h are $\text{W/m}^2\text{K}$.

Rewriting the Equation (1.7.2) in the form,

$$Q = \frac{(T_w - T_\infty)}{\left(\frac{1}{h \cdot A}\right)} \quad \dots(1.7.3)$$

Convective thermal resistance,

$$R = \frac{T_w - T_\infty}{Q} = \frac{1}{h \cdot A} \quad \dots(1.7.4)$$

Where, $\frac{1}{hA}$ represents the **thermal resistance**, R_{con} offered by the film due to heat transfer by convection.

The film conductance, h is the function of many variables such as shape, dimensions of the surface, velocity of fluid, density, temperature, viscosity and specific heat of fluid, coefficient of thermal expansion and coefficient of thermal conductivity i.e.

$$h = f(\rho, V, L/D, \mu, C_p, k)$$

The range of convective heat transfer coefficient for liquids and gases is given in Table 1.7.1

Table 1.7.1 : Coefficient of convective heat transfer, h ($\text{W/m}^2\text{K}$)

Process	h ($\text{W/m}^2\text{K}$)
1. Natural convection	
(i) Still gases	4 to 25
(ii) Still liquids	100 to 900
2. Forced convection	
(i) Air	25 to 100
(ii) Liquids	100 to 3500
(iii) Boiling liquids	2000 to 25000
(iv) Condensation of vapours	4000 to 25000



1.8 Radiation

All bodies radiate heat. **Radiation** is the rate of heat transfer by means of electromagnetic waves that are emitted in the wavelength band between 0.1 and 100 micrometer solely as a result of the temperature of its surface. Thermal radiation being the propagation of electromagnetic waves, it does not require any medium for this propagation.

If two bodies are at different temperatures, the hotter body sends radiations to colder body and the colder body sends radiations to hotter body. As a result the hot body gets cooled and the cold body is heated.

The intensity of radiations emitted by a body depends on the nature of the body and its temperature.

Out of the total radiations falling on a body, a part of it is **reflected** at the surface. (the fraction being indicated **reflectivity, ρ**) a part of it is absorbed while travelling along the depth of the body and produces heat (denoted by absorptivity, α) and the remainder radiant energy is transmitted through the body (denoted by transmissivity, γ)

$$\text{Therefore, } \rho + \alpha + \gamma = 1 \quad \dots(1.8.1)$$

A body which absorbs entire radiations is called a **black body**. In this case, $\rho = \gamma = 0$, hence $\alpha = 1$. **A black body is also best radiator.**

The bodies which are opaque (usually, solids), the radiant energy falling on it is either absorbed or reflected. In this case $\gamma = 0$, therefore, $\alpha + \rho = 1$.

In certain materials like gases and glass, the entire radiant energy falling on them is transmitted. In this case, $\alpha = \rho = 0$ and $\gamma = 1$.

The body which reflects the entire radiant energy is called the **white body**. For such bodies, $\alpha = \gamma = 0$, thus $\rho = 1$. Such bodies are also called as **diathermanous bodies**.

1.8.1 Emissive Power (E)

The emissive power of any surface is defined as the energy emitted by the surface per unit area per unit time. Its unit are W/m^2 .

1.8.2 Emissivity (ϵ)

Emissivity is defined as the ratio of emissive power of any surface to the emissive power of a black surface at the same temperature. The value of **emissivity ranges between 0 to 1**.

1.9 Laws of Radiation

1. Wien's Law ; 2. Kirchoff's Law
3. Stefan-Boltzmann Law

1.9.1 Wien's Law

It states that the wavelength, λ_m corresponding to maximum energy is inversely proportional to the absolute temperature of the hot body.

$$\text{Therefore } \lambda_m \propto \frac{1}{T}$$

$$\text{i.e. } \lambda_m \cdot T = \text{constant} = 0.0029 \text{ mK} \quad \dots(1.9.1)$$

1.9.2 Kirchoff's Law

It states that the emissivity of the body at a particular temperature is equal to its absorptivity from the body at the same temperature.

1.9.3 Stefan-Boltzmann Law

This law states that the emissive power of a black body is directly proportional to fourth power of its absolute temperature i.e.

$$q \propto T^4$$

$$\text{For area A, } Q \propto A \cdot T^4$$

$$\text{or, } Q = \sigma \cdot A \cdot T^4 \quad \dots(1.9.2)$$

where, q = Emissive power or heat flux (W/m^2)

A = Radiating area (m^2)

Q = Rate of heat energy radiated (W)

σ = Stefan-Boltzmann's constant

$$= 5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$$

Equation (1.9.2) holds good for **black body** only.

A body having emissivity less than 1 and it is same for all wavelength is called gray body. Its absorptivity is equal to emissivity.

For a gray body

$$Q = \epsilon \cdot \sigma \cdot A \cdot T^4 \quad \dots(1.9.3)$$

where, ϵ is called emissivity as defined in section 1.8.2.



Now consider a body of surface area A at temperature T_1 kept in infinite surroundings at temperature T_2 ($T_1 > T_2$). The exchange of energy by radiation from a solid to its surroundings can be written as :

$$Q = \epsilon \cdot \sigma \cdot A (T_1^4 - T_2^4) \quad \dots(1.9.4)$$

Also, $Q = \epsilon \cdot \sigma \cdot A (T_1^2 - T_2^2) (T_1 + T_2) (T_1 - T_2)$

$$\therefore Q = \frac{\epsilon \cdot \sigma \cdot A (T_1^2 - T_2^2) (T_1 + T_2)}{1/[\epsilon \cdot \sigma \cdot A (T_1^2 + T_2^2) (T_1 + T_2)]} \quad \dots(1.9.5)$$

In the above equation, the denominator represents the radiation thermal resistance.

Note : The emissivity for black body, $\epsilon = 1$.

1.10 General Heat Conduction Equation in Cartesian Co-ordinates

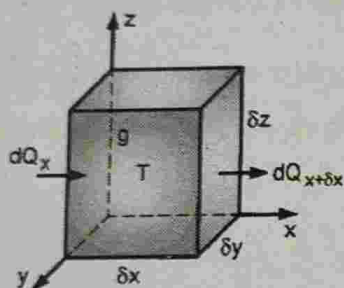


Fig. 1.10.1 : Volume element for determining general heat conduction equation

Refer Fig. 1.10.1.

Consider a small rectangular volume of sides δx , δy and δz parallel to the three axes in medium, in which temperature is varying with location and time.

Let T denote the temperature at centre of this elemental volume.

Also, let there be internal heat generation at the rate of g watt per unit volume (W/m^3) due to heat source.

Let the material be **anisotropic** implying that thermal conductivities have values k_x , k_y and k_z in x , y and z directions respectively.

Consider that heat is entering and leaving this volume through its six faces.

Consider the heat flow in X -direction : Heat entering the elemental volume per unit time normal to the area/face $\delta y \delta z$ at ' x ' be dQ_x and heat leaving the volume from opposite face $\delta y \delta z$ at $(x + \delta x)$ be $dQ_{x+\delta x}$.

As per Fourier's law of heat conduction $Q = -kA \frac{dT}{dx}$

Heat entering, $dQ_x = -k_x (\delta y \delta z) \partial T / \partial x$

The value of heat flow out of the element from right face can be obtained by expanding dQ_x in Taylor's series and retaining the first two terms only.

$$\therefore \text{Heat leaving, } dQ_{x+\delta x} = dQ_x + \partial / \partial x (dQ_x) \delta x +$$

...So, net heat flow into the element in X -direction/time,

$$\begin{aligned} dQ_x - dQ_{x+\delta x} &= -\frac{\partial}{\partial x} (dQ_x) \delta x = -\frac{\partial}{\partial x} \left(-k_x \delta y \delta z \cdot \frac{\partial T}{\partial x} \right) \delta x \\ &= \frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) \delta x \delta y \delta z \quad \dots(i) \end{aligned}$$

Similarly, net heat flow into the element per unit time in y and z directions respectively are :

$$dQ_y - dQ_{y+\delta y} = \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) \delta x \delta y \delta z \quad \dots(ii)$$

$$dQ_z - dQ_{z+\delta z} = \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) \delta x \delta y \delta z \quad \dots(iii)$$

* Thus, net heat flow in to the element from all directions by conduction in certain time δt will be,

$$\left[\frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) \right] \delta x \delta y \delta z \delta t \quad \dots(iv)$$

Now consider that there is some heat source within the element which generates heat given as :

g = heat generated per unit volume per unit time

$$\text{Volume} = \delta x \cdot \delta y \cdot \delta z$$

* Therefore, internal heat generation in time δt .

$$= g \cdot \delta x \delta y \delta z \delta t \quad \dots(v)$$

Heat gained by the element from above will result into energy storage and will increase its temperature.

Let, C_p = Specific heat of element material

ρ = density

δT = Rise in temperature in time δt

* The net heat storage in the element in time δt will be,

$$\begin{aligned} (m C_p \Delta T) &= (\rho V) C_p \delta T \\ &= \rho C_p \delta T \delta x \delta y \delta z \quad \dots(vi) \end{aligned}$$

Energy Balance Equation

Net heat conducted in to the element from all directions + Heat generated within the element

= Energy stored in the element.

$$\left[\frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) \right] \delta x \delta y \delta z \delta t + g \cdot \delta x \delta y \delta z \delta t = \rho C_p \delta T \cdot \delta x \delta y \delta z \quad \dots(1.10.1)$$

Dividing the equation by $\delta x \delta y \delta z \cdot \delta t$ we get,

$$\therefore \frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) + g = \rho C_p \frac{\partial T}{\partial t} \quad \dots(1.10.2)$$

Equations (1.10.2) is the three dimensional heat conduction equations in Cartesian co-ordinates for anisotropic material under unsteady state conditions.

For isotropic material, $k_x = k_y = k_z = k$ constant.

Equation (1.10.2) reduces to,

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{g}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \dots(1.10.3)$$

(where, $\frac{1}{\alpha} = \frac{\rho C_p}{k}$)

Where, α represents the thermal diffusivity of the material. It is given as :

$$\text{Thermal diffusivity, } \alpha = \frac{k}{\rho \cdot C_p} \quad \dots(1.10.4)$$

Significance of Thermal diffusivity

The units of thermal diffusivity, α are m^2/s . It indicates that larger the value of α , faster will be heat diffusion through a material.

The thermal conductivity of metal is many times greater than that of gases but the density of gases is much smaller than metals, for this reason the heat diffuses through gases almost at the same rate as it passes through the metals.

1.11 Special Cases of General Heat Conduction Equation

Equation (1.10.2) represents the general heat conduction equation under unsteady state for **anisotropic materials** and Equation (1.10.3) represents the equation for **isotropic materials**

(i.e. $k = k_x = k_y = k_z$).

This equation can be reduced for certain special cases as follows :

- (i) Steady state conduction
- (ii) No heat sources
- (iii) No heat source and steady state conditions
- (iv) One dimensional heat conduction equation without heat generation under steady state

(i) Steady state conduction

The system is said to be in steady state if the temperature of material at any point does not change with time i.e. $\frac{\partial T}{\partial t} = 0$

The heat conduction Equation (1.10.3) becomes :

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{g}{k} = 0 \quad [\text{Poisson's Equation}] \quad \dots(1.11.1)$$

(ii) No heat sources

In absence of any heat generation or release of energy within the body i.e. $g = 0$, the Equation (1.10.3) reduces to :

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad [\text{Fourier's Equation}] \quad \dots(1.11.2)$$

(iii) No heat source and steady state conditions

In this case, $g = 0$ (no heat source) and $\frac{\partial T}{\partial t} = 0$ (steady state), the Equation (1.10.3) reduces to :

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0 \quad [\text{Laplace Equation}] \quad \dots(1.11.3)$$

$$\text{or } \nabla^2 T = 0 \quad \dots(1.11.4)$$

(iv) One dimensional heat conduction equation without heat generation under steady state

Consider the heat flow in X-direction only with no heat generation ($g = 0$) and under steady state ($\frac{\partial T}{\partial t} = 0$).

The Laplace Equation (1.11.3) reduces to :

$$\frac{\partial^2 T}{\partial x^2} = 0 \quad \text{or} \quad \frac{d^2 T}{dx^2} = 0 \quad \dots(1.11.5)$$

1.12 Heat Conduction Equation in Cylindrical Co-ordinates

Consider an element having polar co-ordinates (r, θ, z) . Three sides are dr , dz and $r \cdot d\theta$ as shown in Fig. 1.12.1. By substituting $x = r \cdot \cos \theta$, $y = r \cdot \sin \theta$ and $z = z$, the three dimensional heat conduction. Equation (1.10.3) reduces to :

$$\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \left(\frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \cdot \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} + \frac{g}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \dots(1.12.1)$$

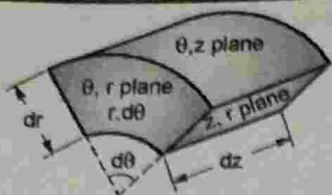


Fig. 1.12.1 : Elemental volume in cylindrical co-ordinate for three dimensional heat conduction equation

1.12.1 Steady State One Dimensional Heat Conduction Equation (Poisson's Equation)

With $\frac{\partial T}{\partial t} = 0$ (Steady state)

Equation (1.12.1) for one dimensional heat conduction for isotropic material reduces to :

$$\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \cdot \frac{\partial T}{\partial r} + \frac{g}{k} = 0 \quad \dots(1.12.2)$$

$$\frac{1}{r} \cdot \frac{\partial}{\partial r} \left(r \cdot \frac{\partial T}{\partial r} \right) + \frac{g}{k} = 0 \quad \dots(1.12.3)$$

In case no heat generation i.e. $g = 0$, Equation (1.12.3) reduces to :

$$\frac{1}{r} \cdot \frac{\partial}{\partial r} \left(r \cdot \frac{\partial T}{\partial r} \right) = 0 \quad \dots(1.12.4)$$

1.13 General Heat Conduction Equation in Spherical Co-ordinates

Consider a cylindrical differential element of isotropic material as shown in Fig. 1.13.1. It's co-ordinates are (r, θ, ϕ) . By substituting

$$x = r \cdot \sin \theta \cdot \cos \phi,$$

$$y = r \cdot \sin \theta \cdot \sin \phi \text{ and}$$

$$z = r \sin \theta$$

in Equation (1.10.3) we get,

$$\begin{aligned} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \cdot \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T}{\partial \theta} \right) \\ + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} + \frac{g}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \end{aligned} \quad \dots(1.13.1)$$

For isotropic material with $k = \text{constant}$.

Poisson's Equation :

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) + \frac{g}{k} = 0 \quad \dots(1.13.2)$$

Equation (1.13.2) is applicable to one dimensional, steady state heat conduction with heat generation.

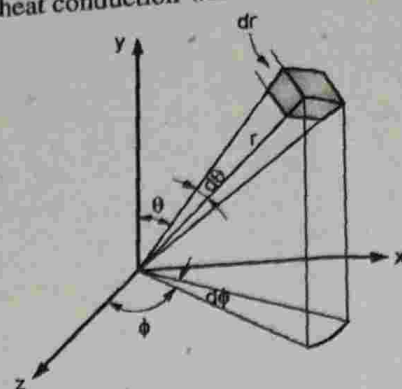


Fig. 1.13.1 : Element for general heat conduction equation in spherical co-ordinates

Section II : Boundary and Initial Conditions

1.14 Boundary and Initial Conditions

There will be the requirement of determining temperature distribution through a solid or fluid to design thermal systems. To work out this requirement, some physical conditions existing at the boundaries (faces) of the thermal systems should be known or measurable. Minimum two conditions would be required in any one direction of heat transfer, which are under steady state conditions and minimum one condition shall be necessary at the initial stage of time for thermal systems under transient condition of heat transfer. The two conditions specified for systems under steady state heat transfer, are called boundary conditions as there are specified at boundaries (faces) of the system. When a system is under transient conditions, the condition specified shall be at the commencement of heat transfer and is known as initial conditions.

1.15 Types of Boundary Conditions

There are generally four types of boundary conditions. These are :

- (1) Temperature boundary conditions
- (2) Convection boundary conditions
- (3) Flux boundary conditions
- (4) Initial boundary conditions

1.15.1. Temperature Boundary Conditions

If the heat conduction is taking place across a wall of thickness L (Fig. 1.15.1), then to find out heat flow rate and temperature distribution through wall thickness L , we should be able to either measure the temperatures T_1 and T_2 of isothermal faces or these should be specified.

Hence these boundary conditions can be written for x -direction as :

- (1) At $x = 0$, $T = T_1$ (2) At $x = L$, $T = T_2$

Time t is not applicable to steady state condition.

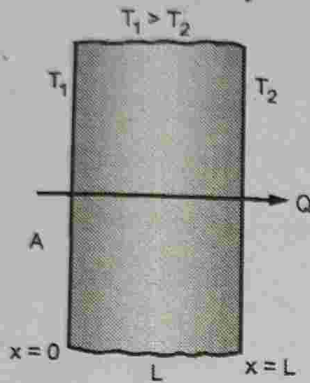


Fig. 1.15.1

1.15.2 Convection Boundary Conditions

When face temperatures (T_1 and T_2) are not known/ specified, convection boundary conditions are to be considered. In this case, fluid temperature $T_{\infty 1}$, $T_{\infty 2}$, and convection heat transfer coefficients h_1 and h_2 are supposed to be known. In this case, boundary conditions can be written as,

- (1) At $x = 0$,
Heat convected into wall = Heat conducted into wall

$$h_1 A (T_{\infty 1} - T_1)_{x=0} = -kA \left(\frac{dT}{dx} \right)_{x=0}$$

- (2) At right face $x = L$,
Heat conducted upto face = heat convected out from face
 $-kA \left(\frac{dT}{dx} \right)_{x=L} = h_2 A (T_2 - T_{\infty 2})_{x=L}$

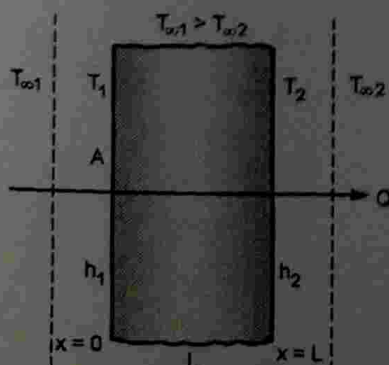


Fig. 1.15.2

1.15.3 Heat Flux Boundary Conditions

Heat flux is given as,

$$q_x = \frac{Q_x}{A} = -k \frac{dT}{dx}$$

Two boundary conditions can be specified as :

- (1) At left face, $x = 0$

Heat flux entering the wall,

$$q_0 = -k \left(\frac{dT}{dx} \right)_{x=0}$$

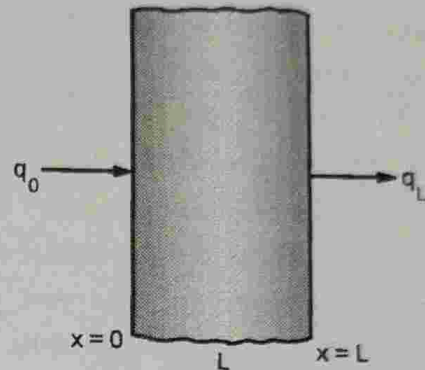


Fig. 1.15.3

- (2) At right face, $x = L$

Heat flux leaving the wall,

$$q_L = -k \left(\frac{dT}{dx} \right)_{x=L}$$

Suppose, one face, say $x = 0$, is insulated,

Then flux $q_0 = 0$

$$\therefore -k \left(\frac{dT}{dx} \right)_{x=0} = 0 \quad \therefore \left(\frac{dT}{dx} \right)_{x=0} = 0$$

1.15.4 Initial Boundary Conditions

Initial boundary conditions are applicable to transient heat transfer, when conditions are specified/ measured at the commencement of heat transfer i.e. at time $t = 0$.

Time is not required to be considered in steady state heat transfer.

Initial boundary condition can be specified, for example, in heat treatment of metal process, where a metal piece has been heated to certain known temperature T_i and it is suddenly placed in fluid at temperature T_{∞} , for quenching ($T_i > T_{\infty}$), then boundary condition

Written as; at time $t = 0$, $T = T_i$.



Section III : One Dimensional Steady State Heat Conduction without Heat Generation

1.16 One Dimensional Steady State Heat Conduction without Heat Generation

Previously we have discussed the three modes of heat transfer. In actual heat transfer problems, the heat transfer usually takes place by all the three modes of heat transfer i.e. by conduction, convection and radiation. In case the heat conduction is considered in one direction only with negligible heat transfer in other directions, the problem is considered as one dimensional heat transfer problem.

We shall now discuss the problem of rate of heat transfer as combined modes of heat transfer under steady state (i.e. the temperature within the system at any point remains constant and it is invariant with time) without heat generation within the system.

Problems associated with combined heat transfer can easily be solved by using the concepts of electrical analogy and the overall heat transfer coefficient.

1.17 Analysis of One Dimensional Steady State Heat Conduction without Internal Heat Generation

The cases in which there is no internal heat generation have already been dealt in previous sections. However, we shall now verify those results by using the differential equations derived above.

1.17.1 One Dimensional Steady State Heat Conduction through Plane Wall or Infinite Slab without Heat Generation

Consider a plane wall or infinite slab of thickness 'L' of material having thermal conductivity k with its faces maintained at T_1 and T_2 .

One dimensional heat conduction equation [Using Laplace Equation (1.11.3) with heat conduction in X-direction only] can be written as :

$$\frac{d^2T}{dx^2} = 0 \quad \dots(i)$$

On integration, $\frac{dT}{dx} = C_1 \quad \dots(ii)$

On integrating again, $T = C_1 x + C_2 \quad \dots(iii)$

Equation (ii) represents the *slope of temperature profile* i.e. slope is constant and Equation (iii) represents the *temperature profile which is linear*.

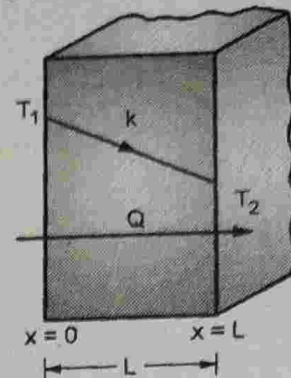


Fig. 1.17.1 : Infinite slab

Boundary conditions are : (Refer Fig. 1.17.1)

- (a) At $x = 0$, $T = T_1$ and
- (b) At $x = L$, $T = T_2$

Using boundary condition at $x = 0$, $T = T_1$ in Equation (iii) we get

$$T_1 = C_1 \times 0 + C_2 \text{ i.e. } C_2 = T_1 \quad \dots(iv)$$

Applying boundary condition at $x = L$, $T = T_2$ in Equation (iii),

$$T_2 = C_1 \times L + C_2 = C_1 \times L + T_1$$

$$C_1 = \frac{T_2 - T_1}{L} \quad \dots(v)$$

On substituting the values of C_1 and C_2 in Equation (iii) we get,

$$T = \left(\frac{T_2 - T_1}{L} \right) \cdot x + T_1$$

$$\text{i.e. } \frac{T - T_1}{T_2 - T_1} = \frac{x}{L}$$

This equation gives the temperature distribution in the slab. From Fourier's law of heat conduction,

$$Q = -k \cdot A \cdot \frac{dT}{dx}$$

But $\frac{dT}{dx} = C_1$ from Equation (ii). On substituting the value of C_1 from Equation (v) we get,

$$\frac{dT}{dx} = C_1 = \frac{T_2 - T_1}{L}$$

Hence, $Q = -k \cdot A \cdot \left(\frac{T_2 - T_1}{L} \right)$
 $= k \cdot A \cdot \frac{(T_1 - T_2)}{L} \quad \dots(1.17.1)$

∴ Thermal resistance,

$$R = \frac{L}{kA} \quad \dots(1.17.2)$$

Heat flux, $\frac{Q}{A} = q = \frac{k(T_1 - T_2)}{L} \quad \dots(1.17.3)$

Above Equations (1.17.1) and (1.17.3) are same as obtained by using Fourier's law of heat conduction in the previous chapter.

1.17.2 One Dimensional (Radial) Steady State Heat Conduction through Hollow Cylinder without Heat Generation

Refer Fig. 1.17.2

Consider a hollow cylinder of inner radius r_1 and outer r_2 of length L of a material having conductivity k .

Inner surface of cylinder is at temperature T_1 and outer at T_2 .

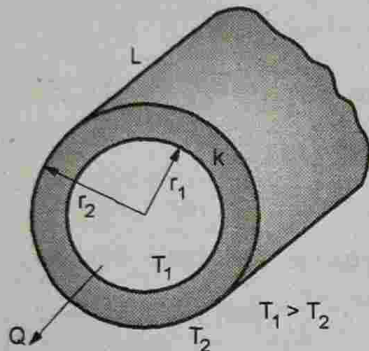


Fig. 1.17.2 : Heat conduction through hollow cylinder

Conduction equation for one dimensional (radial) Heat flow (without g) will be :

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0 \quad [\text{Refer Equation (1.12.4)}] \dots(i)$$

$$\text{i.e.} \quad \frac{d}{dr} \left(r \cdot \frac{dT}{dr} \right) = 0 \quad \dots(ii)$$

On integrating the above equation with respect to 'r'

$$r \cdot \frac{dT}{dr} = C_1, \quad \frac{dT}{dr} = \frac{C_1}{r} \quad \dots(iii)$$

On integrating the above equation again,

$$T = C_1 \ln r + C_2 \quad \dots(iv)$$

Boundary conditions are :

(a) At $r=r_1$, $T=T_1$ (b) At $r=r_2$, $T=T_2$

On substituting the above boundary conditions in Equation (iv) we get,

$$T_1 = C_1 \ln r_1 + C_2 \quad \dots(v)$$

$$T_2 = C_1 \ln r_2 + C_2 \quad \dots(vi)$$

On subtracting Equation (v) from Equation (vi),

$$C_1 = \frac{T_2 - T_1}{\ln r_2 - \ln r_1} = \frac{T_2 - T_1}{\ln \left(\frac{r_2}{r_1} \right)} \quad \dots(vii)$$

Using the value of C_1 in Equation (v) we get,

$$T_1 = \frac{(T_2 - T_1)}{\ln \left(\frac{r_2}{r_1} \right)} \times \ln r_1 + C_2$$

$$\therefore C_2 = T_1 - \frac{(T_2 - T_1)}{\ln \left(\frac{r_2}{r_1} \right)} \ln r_1 \quad \dots(viii)$$

On substituting the values of C_1 and C_2 in Equation (iv),

$$T = \frac{(T_2 - T_1)}{\ln \left(\frac{r_2}{r_1} \right)} \ln r + \left[T_1 - \frac{(T_2 - T_1)}{\ln \left(\frac{r_2}{r_1} \right)} \cdot \ln r_1 \right]$$

$$T = \frac{T_1 - T_2}{\ln \left(\frac{r_1}{r_2} \right)} \ln r + T_1 - \frac{(T_1 - T_2)}{\ln \left(\frac{r_1}{r_2} \right)} \cdot \ln r_1$$

$$\therefore \frac{T - T_1}{T_1 - T_2} = \frac{\ln \left(\frac{r}{r_1} \right)}{\ln \left(\frac{r_1}{r_2} \right)} \quad \dots(1.17.4)$$

Equation (1.17.4) gives the *temperature distribution in hollow cylinder in radial direction without heat generation.*

From Fourier's law of heat conduction,

$$Q = -k \cdot A \cdot \frac{dT}{dr} \quad \text{where } A = 2\pi r L$$

$$\frac{dT}{dr} = \frac{C_1}{r} \quad [\text{From Equation (iii)}]$$

$$\therefore Q = -k \cdot 2\pi r L \cdot \frac{C_1}{r} = -2\pi k L \cdot C_1$$

On substituting the value of C_1 ,

$$\begin{aligned} Q &= -2\pi k \cdot L \cdot \frac{(T_2 - T_1)}{\ln \left(\frac{r_2}{r_1} \right)} \\ &= 2\pi k L \frac{(T_1 - T_2)}{\ln \left(\frac{r_2}{r_1} \right)} = \frac{\Delta T}{R} \quad \dots(1.17.5) \end{aligned}$$

Hence, thermal resistance for hollow cylinder

$$R = \frac{\ln \left(\frac{r_2}{r_1} \right)}{2\pi k L} \quad \dots(1.17.6)$$



1.17.2.1 Logarithmic Mean Area (LMA) for Hollow Cylinder

In Fourier law of heat conduction equation is,

$$Q = -k \cdot A \cdot \frac{dT}{dr}$$

In this equation the heat transfer area, $A = 2\pi r \cdot L$ changes with change in radius r unlike in case of infinite slab where heat transfer area remains constant.

Therefore, it is convenient to work out the mean area A_m for use in analogous formulae for slab,

$$Q = k \cdot A \left(\frac{\Delta T}{\Delta x} \right) \quad \text{Rewriting Equation (1.17.5),}$$

$$Q = \frac{2\pi k L \cdot \Delta T}{\ln \left(\frac{r_2}{r_1} \right)} = k \cdot A_m \cdot \frac{\Delta T}{(r_2 - r_1)} \quad \dots(1.17.7)$$

Where, A_m is the mean area which can be utilized in the formulae for slab.

Multiplying and dividing the Equation (1.17.7) by $(r_2 - r_1)$ we get,

$$\begin{aligned} Q &= \frac{2\pi k L}{\ln \left(\frac{r_2}{r_1} \right)} \cdot \Delta T \cdot \frac{(r_2 - r_1)}{(r_2 - r_1)} \\ &= \frac{k \cdot 2\pi L (r_2 - r_1)}{\ln \left(\frac{r_2}{r_1} \right)} \cdot \frac{\Delta T}{(r_2 - r_1)} \end{aligned}$$

Comparing the above equation with

$$Q = k \cdot A_m \cdot \frac{\Delta T}{(r_2 - r_1)} \text{ we have,}$$

$$A_m = \frac{2\pi L (r_2 - r_1)}{\ln \left(\frac{r_2}{r_1} \right)} = \frac{A_o - A_i}{\ln \left(\frac{A_o}{A_i} \right)} \quad \dots(1.17.8)$$

1.17.3 One Dimensional (Radial) Steady State Heat Conduction through Hollow Sphere without Heat Generation

Refer Fig. 1.17.3, consider a hollow sphere of inner radius r_1 and outer r_2 of a material having conductivity k .

Inner surface of sphere is at temperature T_1 and outer at T_2 .

Conduction equation for one dimensional (radial) Heat flow (without g) will be (Refer Equation (1.13.2))

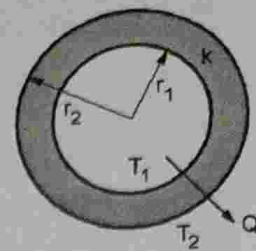


Fig. 1.17.3 : Heat conduction through hollow sphere

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0 \quad \dots(i)$$

$$\text{i.e.} \quad \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0 \quad \dots(ii)$$

$$\text{On integrating,} \quad r^2 \cdot \frac{dT}{dr} = C_1$$

$$\frac{dT}{dr} = \frac{C_1}{r^2} \quad \dots(iii)$$

$$\text{On further integrating,} \quad T = -\frac{C_1}{r} + C_2 \quad \dots(iv)$$

Boundary conditions are

$$(a) \text{ At } r = r_1, T = T_1 \quad (b) \text{ At } r = r_2, T = T_2$$

On substituting the above boundary conditions in Equation (iv) we get,

$$T_1 = -\frac{C_1}{r_1} + C_2 \quad \dots(a)$$

$$T_2 = -\frac{C_1}{r_2} + C_2 \quad \dots(b)$$

On solving above Equations (a) and (b) we get,

$$C_1 = \frac{(T_1 - T_2) r_1 \cdot r_2}{(r_1 - r_2)} \quad \dots(v)$$

$$C_2 = T_1 + \frac{(T_1 - T_2) r_2}{(r_1 - r_2)} \quad \dots(vi)$$

On substituting the values of C_1 and C_2 in Equation (iv) we get,

$$T = -\frac{1}{r} \frac{(T_1 - T_2) r_1 \cdot r_2}{(r_1 - r_2)} + T_1 + \frac{(T_1 - T_2)}{(r_1 - r_2)} \cdot r_2$$

$$T = \frac{r_1}{r} \times \frac{(r_2 - r_1)}{(r_2 - r_1)} \cdot T_1 + \frac{r_2}{r} \times \frac{(r - r_1)}{(r_2 - r_1)} \cdot T_2 \quad \dots(1.17.9)$$

Equation (1.17.9) represents the temperature profile across the thickness of sphere.

From Fourier's law of heat conduction,

$$Q = -k \cdot A \cdot \frac{dT}{dr} = -k \cdot 4\pi r^2 \cdot \frac{dT}{dr}$$

But, $\frac{dT}{dr} = \frac{C_1}{r^2}$, therefore

$$Q = -k \cdot 4\pi r^2 \cdot \frac{C_1}{r^2} = -4\pi k C_1$$

On substituting the value of C_1 ,

$$Q = 4\pi k r_1 \cdot r_2 \cdot \frac{(T_1 - T_2)}{(r_2 - r_1)} = \frac{(T_1 - T_2)}{\left[\frac{(r_2 - r_1)}{4\pi k r_1 r_2} \right]} \quad \dots(1.17.10)$$

$\therefore$ Thermal resistance,

$$R = \frac{r_2 - r_1}{4\pi k r_1 \cdot r_2} \text{ and}$$

$$A_m = 4\pi r_1 \cdot r_2 \quad \dots(1.17.11)$$

1.18 Thermal Diffusivity, α

It is an important property of materials. It is defined as the ratio of thermal conductivity k of the material to heat capacity, $\rho \cdot c$. Accordingly,

Thermal diffusivity,

$$\alpha = \frac{\text{Thermal conductivity, } k}{\text{Heat capacity, } \rho \cdot c} \quad \dots(1.18.1)$$

where, ρ = density of material;

c = specific heat of material

Higher is the value of k , higher is the rate of heat conduction through the material. Whereas, ρc indicates the amount of heat stored per m^3 of material. Thus, the *thermal diffusivity of a material indicates, how fast heat energy propagates through a material.*

1.19 Heat Transfer in an Infinitely Long Hollow Cylinder

- Consider a hollow cylinder of internal radius r_1 and external radius r_2 with respective internal and external temperatures of T_i and T_o as shown in Fig. 1.19.1
- Let, L be the length of cylinder and k is the thermal conductivity of cylinder.
- Heat transfer takes place radially.
- Consider a ring of radius r and thickness dr .

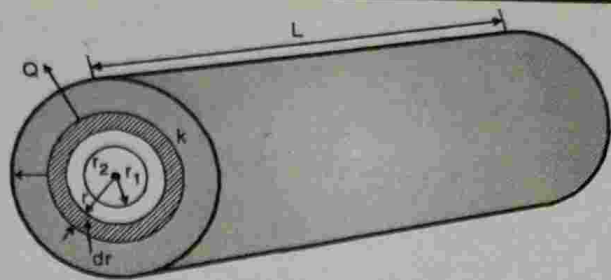


Fig. 1.19.1 : Heat transfer in an infinitely long hollow cylinder

From Fourier's law of heat condition

$$Q = -k \cdot A \cdot \frac{dT}{dx} \quad (\text{But, } A = 2\pi r \cdot L)$$

$$Q = -k \cdot (2\pi r L) \frac{dT}{dr}$$

$$Q \int_{r_1}^{r_2} \frac{dr}{r} = -k \cdot 2\pi \cdot L \int_{T_i}^{T_o} dT$$

$$\begin{aligned} \text{or, } Q &= \frac{2\pi k L (T_i - T_o)}{\log_e \frac{r_2}{r_1}} \\ &= \frac{T_i - T_o}{\frac{\log_e \left(\frac{r_2}{r_1} \right)}{2\pi k L}} = \frac{T_i - T_o}{R} \quad \dots(1.19.1) \end{aligned}$$

Where, $\log_e \left(\frac{r_2}{r_1} \right) / 2\pi k L$ represents the **thermal resistance**.

1.20 Heat Transfer through a Hollow Sphere

Refer Fig. 1.20.1

- Consider a hollow sphere of internal and external radius as r_1 and r_2 respectively with respective temperature T_i and T_o . Heat conduction is in radial direction.
- Consider a ring at radius r of thickness dr .

Surface area of sphere, $A = 4\pi r^2$

$$Q = -k \cdot A \cdot \frac{dT}{dr} = -k \cdot 4\pi r^2 \cdot \frac{dT}{dr}$$

$$\therefore Q \int_{r_1}^{r_2} \frac{dr}{r^2} = -k \cdot 4\pi \int_{T_i}^{T_o} dT$$

$$Q \left[-\frac{1}{r} \right]_{r_1}^{r_2} = -k \cdot 4\pi [T]_{T_i}^{T_o}$$

$$\text{or, } Q = \frac{4\pi k \cdot (T_1 - T_o)}{\left[\frac{1}{r_1} - \frac{1}{r_2}\right]} = \frac{(T_1 - T_o)}{4\pi k \left[\frac{1}{r_1} - \frac{1}{r_2}\right]}$$

$$Q = \frac{T_1 - T_o}{\left(\frac{r_2 - r_1}{4\pi k r_1 r_2}\right)} = \frac{T_1 - T_o}{R} \quad \dots(1.20.1)$$

Hence, $R = \frac{(r_2 - r_1)}{4\pi k r_1 r_2}$ represents the thermal resistance.

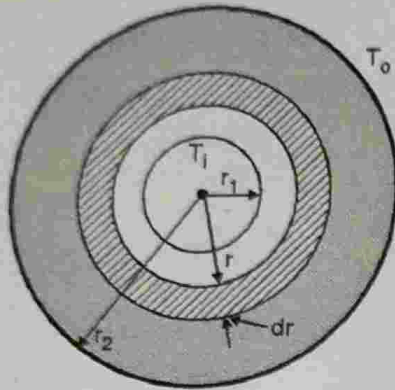


Fig. 1.20.1 : Heat transfer through a hollow sphere

Table 1.20.1 : List of Formulae

$$1. \quad Q = \frac{k \cdot A (T_1 - T_2)}{x} = \frac{(T_1 - T_2)}{(x/kA)}$$

where, $\frac{x}{kA}$ = Conductive resistance, R

$$2. \quad \text{Heat flux, } q = \frac{Q}{A} = \frac{k (T_1 - T_2)}{x}$$

$$3. \quad Q = h \cdot A (T_w - T_\infty) = \frac{(T_w - T_\infty)}{\left(\frac{1}{h \cdot A}\right)}$$

where, $\frac{1}{h \cdot A}$ = Convective resistance, $R_{\text{conv.}}$

$$4. \quad Q = \epsilon \cdot \sigma \cdot A T^4$$

where Stefan Boltzmann's constant

$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$$

$$5. \quad Q = \epsilon \cdot \sigma \cdot A (T_1^4 - T_2^4)$$

$$= \frac{(T_1 - T_2)}{1/\left[\epsilon \cdot \sigma \cdot A (T_1^2 + T_2^2)(T_1 + T_2)\right]} = R_{\text{rad.}}$$

$R_{\text{rad.}}$ = Radiation thermal resistance.

$$6. \quad \text{Fourier's law : } Q = -k \cdot A \frac{dT}{dx}$$

$$7. \quad Q = \frac{k \cdot A \cdot (T_1 - T_2)}{x} = \frac{T_1 - T_2}{\left(R = \frac{x}{kA}\right)}$$

$$8. \quad Q = h \cdot A (T_1 - T_2) = \frac{T_1 - T_2}{\left(\frac{1}{h \cdot A} = R_{\text{conv.}}\right)}$$

9. Rate of heat flow in a hollow cylinder of length L .
($A = 2\pi rL$)

$$Q = \frac{T_1 - T_o}{\ln(r_2/r_1) / 2\pi kL} = \frac{T_1 - T_o}{R}$$

$$\text{where, } R = \frac{\ln(r_2/r_1)}{2\pi kL}$$

10. Hollow Sphere ($A = 4\pi r^2$)

$$Q = \frac{T_1 - T_o}{(r_2 - r_1) / 4\pi k r_1 r_2} = \frac{T_1 - T_o}{R}$$

$$R = \frac{(r_2 - r_1)}{4\pi k r_1 r_2} = \left(\frac{1}{r_1} - \frac{1}{r_2}\right) / 4\pi k$$

1.21 Solved Examples

Ex. 1.21.1 : The wall of a oven is 40 cm thick having its thermal conductivity of 0.7 W/mK. The interior surface of the oven is maintained at a temperature of 800°C and the outside wall temperature is 200°C. The total surface area of wall of the oven is 2 m². Find the thermal resistance, heat flow rate and the heat flux.

Soln. :

Given : Thickness, $x = 40 \text{ cm} = 0.4 \text{ m}$;

Thermal conductivity, $k = 0.7 \text{ W/mK}$;

$T_1 = 800^\circ\text{C}$; $T_2 = 200^\circ\text{C}$; $A = 2 \text{ m}^2$.

(i) Thermal resistance, R

$$R = \frac{x}{k \cdot A} = \frac{0.4}{0.7 \times 2} = 0.2857 \text{ K/W} \quad \dots\text{Ans.}$$

(ii) Heat flow rate, Q

$$Q = \frac{(T_1 - T_2)}{\left(\frac{x}{kA}\right)}$$

$$= \frac{T_1 - T_2}{R}$$

$$= \frac{800 - 200}{0.2857} = 2100 \text{ W} \quad \dots\text{Ans.}$$

(iii) Heat flux, q

$$q = \frac{Q}{A} = \frac{2100}{2} = 1050 \text{ W/m}^2 \quad \dots\text{Ans.}$$

Ex. 1.21.2 : A boiler is made of iron plates 12 mm thick. The temperature of outside surface is 120°C and that of the inner surface is 100°C. Calculate the mass of water evaporated per hour if the heating surface area is 5 m² and thermal conductivity, k for iron is 84 W/mK. Assume enthalpy of evaporation of water at 100°C is 2257 kJ/kg.

Soln. : Refer Fig. P.1.21.2.

Given : Plate thickness, $x = 12 \text{ mm} = 12 \times 10^{-3} \text{ m}$.

Outside temperature, $T_1 = 120^\circ\text{C}$;

Inside temperature, $T_2 = 100^\circ\text{C}$;

$A = 5 \text{ m}^2$; $k = 84 \text{ W/mK}$;

Enthalpy of evaporation $h_{fg} = 2257 \text{ kJ/kg}$.

Heat will flow from outside to inner surface of boiler.

This heat will be utilized to evaporate water inside the boiler at $T_2 = 100^\circ\text{C}$.

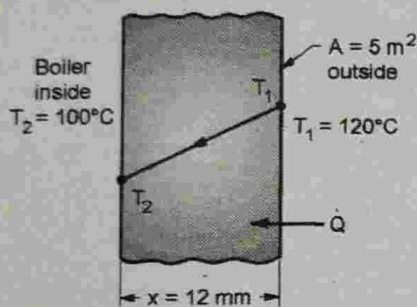


Fig. P. 1.21.2

Heat transfer rate,

$$Q = \frac{kA(T_1 - T_2)}{x} = \frac{84 \times 5 (120 - 100)}{12 \times 10^{-3}}$$

$$= 700 \times 10^3 \text{ W or J/s} = 700 \text{ kJ/s}$$

Let $\dot{m}$ = Rate at evaporation of water .

$$\left\{ \begin{array}{l} \text{Heat utilized by water during evaporation} \\ = \text{Heat transfer rate, } Q \end{array} \right\}$$

$$\dot{m} \times h_{fg} = Q$$

$$\dot{m} = \frac{Q}{h_{fg}} = \frac{700}{2257} \text{ kg/s} = \frac{700}{2257} \times 3600 \text{ kg/hr}$$

$$= 1116.5 \text{ kg/hr} \quad \dots\text{Ans.}$$

Ex. 1.21.3 : Calculate the heat transfer rate per unit area through a copper plate 0.045 m thick whose one face is maintained at 350°C and the other face at 50°C. Thermal conductivity of copper is 370 W/m°C.

Soln. :

Given : Thickness of plate, $x = 0.045 \text{ m}$;

$T_1 = 350^\circ\text{C}$; $T_2 = 50^\circ\text{C}$; $k = 370 \text{ W/m}^\circ\text{C}$

Heat transfer rate per unit area, $\frac{Q}{A}$

$$Q = \frac{kA(T_1 - T_2)}{x}$$

$$q = \frac{Q}{A} = \frac{370 (350 - 50)}{0.045}$$

$$= 2466.667 \times 10^3 \text{ W/m}^2$$

$$= 2466.667 \text{ kW/m}^2 \quad \dots\text{Ans.}$$

Ex. 1.21.4 : A plane wall is 15 cm thick of surface area 4.5 m². Thermal conductivity of the wall is 9.5 W/mK. The inner and outer surface temperatures of the wall are maintained at 150°C and 45°C respectively. Determine,

- Heat flow rate across the wall.
- Temperature gradient in the heat flow direction and
- Temperature of surfaces at 5 cm and 10 cm away from the inner surface.

Soln. :

Given : Thickness, $x = 15 \text{ cm} = 0.15 \text{ m}$;

$A = 4.5 \text{ m}^2$, $k = 9.5 \text{ W/mK}$;

$T_1 = 150^\circ\text{C}$; $T_2 = 45^\circ\text{C}$.

Refer Fig. P. 1.21.4.

(i) Heat flow rate across the wall, Q

$$Q = \frac{kA(T_1 - T_2)}{x} = \frac{9.5 \times 4.5 (150 - 45)}{0.15}$$

$$= 29925 \text{ W} \quad \dots\text{Ans.}$$

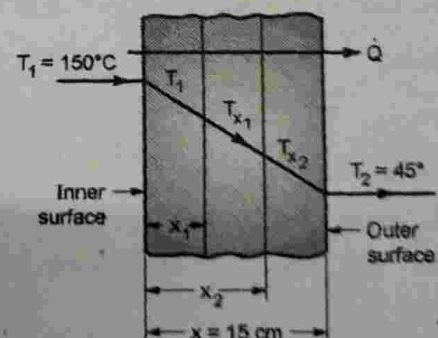


Fig. P. 1.21.4

(ii) Temperature gradient in the heat flow direction, $\frac{dT}{dx}$

From Fourier's law of heat conduction,

$$Q = -kA \frac{dT}{dx}$$

$$\therefore \frac{dT}{dx} = -\frac{Q}{kA} = -\frac{29925}{9.5 \times 4.5}$$

$$= -700^\circ\text{C/m} \quad \dots\text{Ans.}$$

(iii) Temperature of surfaces at $x_1 = 5 \text{ cm} = 0.05 \text{ m}$ and at $x_2 = 10 \text{ cm} = 0.1 \text{ m}$ from inner surface.

Let T_{x1} and T_{x2} be the temperatures at respective surfaces at a distance x_1 and x_2 . Since the heat transfer rate Q is constant throughout, under steady state conditions, it implies that :

$$Q = \frac{kA(T_1 - T_{x1})}{x_1}$$

$$\therefore 29925 = \frac{9.5 \times 4.5 (150 - T_{x1})}{0.05}$$

$$T_{x1} = 115^\circ\text{C} \quad \dots\text{Ans.}$$

$$Q = \frac{kA(T_1 - T_{x2})}{x_2}$$

$$29925 = \frac{9.5 \times 4.5 (150 - T_{x2})}{0.1}$$

$$T_{x2} = 80^\circ\text{C} \quad \dots\text{Ans.}$$

Ex. 1.21.5 : The wall thickness of the oven is 8.25 cm and thermal conductivity of wall insulation is 0.044 W/mK. The temperature of inner surface of the wall is 175°C. The energy dissipated by the electric coil within the oven is 40.5 W. Determine the area of wall surface needed perpendicular to heat flow so that the temperature on the other side of the wall does not exceed 75°C.

Soln. :

Given : Wall thickness, $x = 8.25 \text{ cm} = 0.0825 \text{ m}$;

$k = 0.044 \text{ W/mK}$; $T_1 = 175^\circ\text{C}$;

$Q = 40.5 \text{ W}$; $T_2 = 75^\circ\text{C}$.

Let A be the surface area needed.

Energy dissipated by coil,

Q in oven = Heat transfer rate by conduction across the wall

$$Q = \frac{kA(T_1 - T_2)}{x}$$

$$40.5 = \frac{0.044 A (175 - 75)}{0.0825}$$

$$A = 0.7594 \text{ m}^2 \quad \dots\text{Ans.}$$

Ex. 1.21.6 : A hot plate area 1.5 m^2 is maintained at 300°C . The air at 20°C blows over the plate. If the convective heat transfer coefficient is $20 \text{ W/m}^2\text{K}$, calculate the rate of convective heat transfer.

Soln. :

Given : Plate temperature, $T_w = 300^\circ\text{C}$; $A = 1.5 \text{ m}^2$;

Film temperature (air), $T_\infty = 20^\circ\text{C}$; $h = 20 \text{ W/m}^2\text{K}$.

Rate of convective heat transfer,

$$Q = h \cdot A (T_w - T_\infty)$$

$$= 20 \times 1.5 (300 - 20)$$

$$= 8400 \text{ W} \quad \dots\text{Ans.}$$

Ex. 1.21.7 : An immersion water heater of surface area 0.1 m^2 and rating 1 kW is designed to operate fully submerged in water. Estimate the surface temperature of the heater when the water is at 40°C and $h_{\text{water}} = 300 \text{ W/m}^2\text{K}$. If this heater is by mistake used in air at 40°C with $h_{\text{air}} = 9 \text{ W/m}^2\text{K}$, what will be the surface temperature?

Soln. :

Given : $A = 0.1 \text{ m}^2$, $Q = 1 \text{ kW}$

$T_w = 40^\circ\text{C}$,

$h_w = 300 \text{ W/m}^2\text{K}$

$T_a = 40^\circ\text{C}$,

$h_a = 9 \text{ W/m}^2\text{K}$,

$T_s = ?$

Water : $Q = \frac{\Delta T}{R}$

$$1000 = \frac{T_s - T_w}{\frac{1}{h_w A}} = \frac{T_s - 40}{\frac{1}{300 \times 0.1}}$$

$$T_s = 73.33^\circ\text{C} \quad \dots\text{Ans.}$$

$$\text{Air : } 1000 = \frac{T_s - 40}{\frac{1}{9 \times 0.1}}$$

$$T_s = 1151^\circ\text{C} \quad \dots\text{Ans.}$$

Ex. 1.21.8 : A wire 2.0 mm in diameter and 18 cm long is submerged in water at atmospheric pressure. An electric current is passed through the wire until the water boils at 100°C . In case the wire surface temperature is needed to be maintained at 125°C , determine the electric power supplied to the wire. Assume, convective heat transfer coefficient as $4000 \text{ W/m}^2\text{K}$.

Soln. :

Given : Diameter of wire, $d = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$

Length of wire, $L = 18 \text{ cm} = 0.18 \text{ m}$;

Wire surface temperature, $T_w = 125^\circ\text{C}$;

Water (fluid) temperature, $T_{\infty} = 100^{\circ}\text{C}$;

$$h = 4000 \text{ W/m}^2\text{K}.$$

Surface area of wire exposed, $A = \pi DL$

$$= \pi \times (2 \times 10^{-3}) \times 0.18$$

$$= 1.131 \times 10^{-3} \text{ m}^2$$

Electric power needed to be supplied to wire = Rate of convective heat loss

$$\therefore Q = h \cdot A (T_w - T_{\infty})$$

$$Q = 4000 \times (1.131 \times 10^{-3}) \times (125 - 100)$$

$$= 113.1 \text{ W}$$

...Ans.

Ex. 1.21.9 : A brick wall ($k = 1.5 \text{ W/mK}$) 0.20 m thickness separates hot combustion gases of a furnace from outside ambient air which is at 25°C . The outer surface temperature of brick wall is found to be 100°C . If natural convection heat transfer coefficient on the outside surface of brick wall is $25 \text{ W/m}^2\text{K}$, $\epsilon = 0.85$, calculate inner surface temperature of brick wall.

Soln. :

Given : $k = 1.5 \text{ W/mK}$

$\Delta x = 0.20 \text{ m}$; $T_{\infty} = 25^{\circ}\text{C}$; $T_o = 100^{\circ}\text{C}$

$h = 25 \text{ W/m}^2\text{K}$; $\epsilon = 0.85$; $T_i = ?$

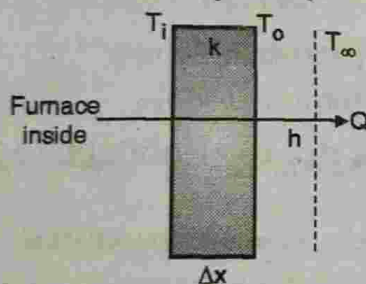


Fig. P. 1.21.9

Heat flow rate

$$Q = Q_c + Q_r$$

$$Q_c = \frac{T_o - T_{\infty}}{\frac{1}{hA}} = \frac{100 - 25}{\frac{1}{25 \times 1}} = 1875 \text{ W/m}^2$$

$$\begin{aligned} Q_r &= \epsilon \sigma A (T_o^4 - T_{\infty}^4) \\ &= 0.85 \times 5.67 \times 10^{-8} \times 1 [(100 + 273)^4 - (25 + 273)^4] \\ &= 4.8195 \times 10^{-8} (193.569 \times 10^8 - 78.86 \times 10^8) = 552.84 \text{ W/m}^2 \end{aligned}$$

Total heat transfer rate

$$Q = 1875 + 552.84 = 2427.85 \text{ W/m}^2$$

$$\text{Now, } Q = 2427.85 = \frac{T_i - 100}{\frac{\Delta x}{kA}} = \frac{T_i - 100}{\frac{0.2}{1.5 \times 1}}$$

$$T_i = 423.7^{\circ}\text{C}$$

...Ans.

Ex. 1.21.10 : A horizontal steel pipe having a diameter of 5 cm is maintained at a temperature of 50°C in a large room where the air and wall temperature are at 20°C . The surface emissivity of the steel may be taken as 0.8 and the convection heat transfer coefficient as $25 \text{ W/m}^2\text{K}$. Calculate the total heat lost by the pipe per unit length.

Soln. :

Given : $T_s = 50^{\circ}\text{C}$, $d = 5 \text{ cm} = 0.05 \text{ m}$;

$$\epsilon_s = 0.8, \quad h = 25 \text{ W/m}^2\text{K}$$

Heat transfer by convection :

$$Q_c = h A \Delta T$$

$$= 25 \times \pi \times 0.05 \times 1 \times (50 - 20) = 117.75 \text{ W}$$

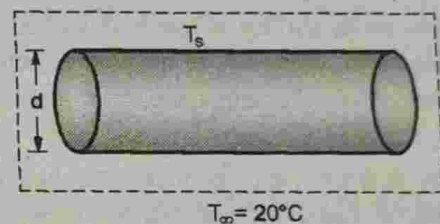


Fig. P. 1.21.10

Heat loss by radiation :

$$Q_r = \epsilon A (T_s^4 - T_{\infty}^4)$$

$$\begin{aligned} &= 5.67 \times 10^{-8} \times 0.8 \times \pi \times 0.05 \times 1 \\ &\times [(50 + 273)^4 - (20 + 273)^4] = 25 \text{ W} \end{aligned}$$

Total heat loss rate from pipe per meter :

$$Q = Q_c + Q_r$$

$$= 117.75 + 25 = 142.75 \text{ W}$$

...Ans.

Ex. 1.21.11 : Air at 27°C blows over a hot plate of $0.5 \text{ m} \times 1 \text{ m}$ surface which is maintained at 227°C . The film conductance is $25 \text{ W/m}^2\text{K}$. There is a heat loss of 280 W by radiation from surface of the plate. The plate is 2 cm thick.

Calculate :

(i) Heat transfer rate.

(ii) The temperature of the other side of the plate.

Assume thermal conductivity of plate material as 43 W/mK .

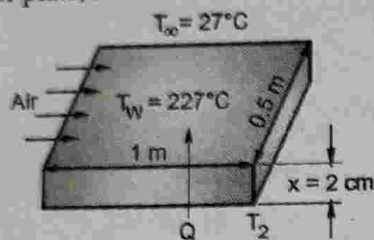
**Soln. :****Given :** Refer Fig. P. 1.21.11.Heat loss by radiation, $Q_r = 280 \text{ W}$ Air (fluid) temperature, $T_\infty = 27^\circ\text{C}$ Surface temperature, $T_w = 227^\circ\text{C}$ Film conductance, $h = 25 \text{ W/m}^2\text{K}$,Thickness of plate, $x = 2 \text{ cm} = 0.02 \text{ m}$; $k = 43 \text{ W/mK}$,Surface area of plate, $A = 1 \times 0.5 = 0.5 \text{ m}^2$.

Fig. P. 1.21.11

(i) Heat Transfer rate :

Heat from upper surface is lost both by radiation and convection. However, other side of the plate is being heated so that the upper surface temperature of plate could be maintained at $T_w = 227^\circ\text{C}$.

It implies that the heat conducted through the plate thickness must be equal to heat lost by convection and radiation.

Heat lost by convection,

$$Q_c = h \cdot A (T_w - T_\infty)$$

$$= 25 \times 0.5 \times (227 - 27)$$

$$Q_c = 2500 \text{ W}$$

 $\therefore$ Total heat lost or heat transfer rate,

$$Q = \text{Heat lost by convection, } Q_c$$

$$+ \text{Heat loss by radiation, } Q_r$$

$$= 2500 + 280 = 2780 \text{ W} \quad \dots\text{Ans.}$$

(ii) Temperature of other side of the plate, T_2

$$Q = \frac{k \cdot A \cdot (T_2 - T_w)}{x}$$

$$2780 = \frac{43 \times 0.5 \times (T_2 - 227)}{0.02}$$

$$= 229.586^\circ\text{C} \quad \dots\text{Ans.}$$

Ex. 1.21.12 : Using approximate assumption, calculate the radiation heat transfer coefficient for a small hot surface having temperature of 152°C with emissivity of 0.85 dissipates heat by radiation into a hot surrounding area.

Soln. : Surface temperature, $T_s = 152^\circ\text{C} = 425\text{K}$,Emissivity, $\epsilon = 0.85$ Let surrounding temperature, $T_\infty = 25^\circ\text{C} = 298 \text{ K}$

The heat transfer rate by radiation from body to surroundings,

$$Q = \epsilon \cdot A \cdot \sigma (T_s^4 - T_\infty^4) \quad \dots(i)$$

Let h_r be the radiation heat transfer coefficient similar to convective heat transfer coefficient.

$$\text{Then, } Q = h_r \cdot A (T_s - T_\infty) \quad \dots(ii)$$

On equating Equation (i) and (ii) we get,

$$\epsilon \cdot A \cdot \sigma (T_s^4 - T_\infty^4) = h_r \cdot A (T_s - T_\infty)$$

$$\therefore 0.85 \times A \times (5.67 \times 10^{-8}) (425^4 - 298^4)$$

$$= h_r \times A (425 - 298)$$

$$\therefore h_r = 9.39 \text{ W/m}^2\text{K} \quad \dots\text{Ans.}$$

Ex. 1.21.13 : A horizontal plate ($k = 30 \text{ W/mK}$) $600 \text{ mm} \times 900 \text{ mm} \times 30 \text{ mm}$ is maintained at 300°C . The air at 30°C flows over the plate. If the convection coefficient of air over the plate is $22 \text{ W/m}^2\text{K}$ and 250 W heat is lost from the plate by radiation, calculate the bottom surface temperature of the plate.

Soln. : Area of plate transferring heat

$$A = 600 \text{ mm} \times 900 \text{ mm}$$

$$= 54 \times 10^4 \text{ mm}^2 = \frac{54 \times 10^4}{10^6} \text{ m}^2 = 0.54 \text{ m}^2$$

Thickness of plate, $L = 30 \text{ mm} = 0.03 \text{ m}$

Surface temperature of plate,

 $T_s = 300^\circ\text{C}$; Temperature of air, $T_\infty = 30^\circ\text{C}$ $k = 30 \text{ W/mK}$; $h = 22 \text{ W/m}^2\text{K}$, $Q_{\text{radiation}} = 250 \text{ W}$ Let bottom surface temperature be T_1

In this case the heat is conducted through the plate which is lost from the plate surface by a combination of convection and radiation.

Q conduction = Q convection + Q radiation

$$\frac{k \cdot A (T_1 - T_s)}{L} = h \cdot A (T_s - T_\infty)$$

$$+ Q_{\text{radiation}}$$

$$\therefore \frac{30 \times 0.54 (T_1 - 30)}{0.03} = 22 \times 0.54 (300 - 30) + 250$$

$$\therefore 540 (T_1 - 300) = 3207.6 + 250 = 3457.6$$

$$\therefore T_1 = 306.403^\circ\text{C} \quad \dots\text{Ans.}$$

Ex. 1.21.14 : In a furnace, temperature at hot gases is 2100°C . Ambient temperature is 40°C . Heat flow by radiation from hot gases to inner surface of the wall is 23 kW/m^2 . Convective heat transfer coefficient between hot gases and the inner surface of the wall is $12 \text{ W/m}^2 \text{ K}$. Thermal conductance of the wall is $58 \text{ W/m}^2 \text{ K}$. Heat flow by radiation from external surface of the wall to surroundings is 10 kW/m^2 . Temperature of inside surface of the wall is 900°C . For the external surface of the wall, find surface temperature and convective heat transfer coefficient.

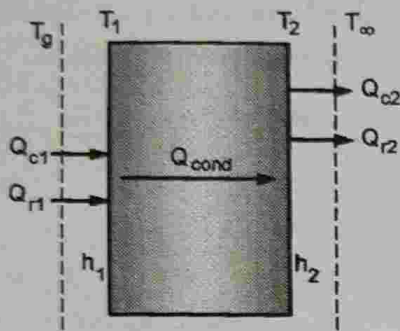


Fig. P. 1.21.14

Soln. : Refer Fig. P. 1.21.14.

Let Q_{c1} and Q_{r1} be the rates of heat transfer by convection and radiation respectively from gases to inner surface of the wall maintained at temperature T_1 . Let T_2 be the temperature on outer surface of the wall from where the rate of heat transfer by convection and radiation be Q_{c2} and Q_{r2} respectively to the surroundings at T_{∞} .

Given : Temperature of hot gases, $T_g = 2100^{\circ}\text{C}$,

$T_{\infty} = 40^{\circ}\text{C}$, $Q_{r1} = 23 \text{ kW/m}^2$, $h_1 = 12 \text{ W/m}^2 \text{ K}$,

Thermal conductance of wall, $\frac{1}{R_c} = 58 \text{ W/m}^2 \text{ K}$,

$Q_{r2} = 10 \text{ kW/m}^2$, $T_1 = 900^{\circ}\text{C}$.

To find : T_2 and h_0 , Assume, $A = 1 \text{ m}^2$.

(i) Temperature at outer surface of wall or slab, T_2

$$\begin{aligned} Q_{c1} &= h_1 \cdot A (T_g - T_1) \\ &= 12 \times 1 \times (2100 - 900) \\ &= 14.4 \times 10^3 \text{ W/m}^2 = 14.4 \text{ kW} \end{aligned}$$

Total heat transfer, Q to inner wall is by radiation and convection. Therefore,

$$Q = Q_{c1} + Q_{r1} = 14.4 + 23 = 37.4 \text{ kW/m}^2$$

This heat is conducted through the slab. Hence,

$$Q = R_c = \frac{T_1 - T_2}{R_c}$$

$$\text{Since } \frac{1}{R_c} = 58 \text{ W/m}^2 \text{ K}$$

$$37.4 \times 10^3 = 58 \times 1 \times (900 - T_2)$$

$$T_2 = 255.2^{\circ}\text{C}$$

...Ans.

(ii) Convective heat transfer coefficient at outer surface, h_0

$$\therefore Q = Q_{r2} + Q_{c2} \text{ as under steady state}$$

Q will remain same.

$$37.4 = 10 + Q_{c2}$$

$$Q_{c2} = 27.4 \text{ kW}$$

$$\text{But, } Q_{c2} = h_0 \cdot A (T_2 - T_{\infty})$$

$$27.4 \times 10^3 = h_0 \times 1 \times (255.2 - 40)$$

$$h_0 = 127.32 \text{ W/m}^2 \text{ K}$$

...Ans.

Ex. 1.21.15 : A brick ($k = 1.2 \text{ W/mK}$) wall 0.15 m thick separates hot combustion gases of a furnace from the outside ambient air which is at 25°C . The outer surface temperature of the brick wall is found to be 100°C . If the natural convection heat transfer coefficient on outside surface of the brick wall is $20 \text{ W/m}^2 \text{ K}$ and its emissivity is 0.8 , calculate the inner surface temperature of the brick wall.

Soln. :

Refer Fig. P. 1.21.15.

Outside wall surface temperature,

$$T_2 = 100^{\circ}\text{C} = 373 \text{ K}$$

Consider unit area of brick wall and

Let T_1 = temperature of wall on gas side.

The rate of heat conducted through the brick wall must be equal to the sum of rate of heat radiated and heat convected from outside surface of the wall to surrounding air i.e.

$$\begin{aligned} q_{\text{cond.}} &= q_{\text{rad.}} + q_{\text{conv.}} \\ \frac{k(T_1 - T_2)}{L} &= \epsilon \cdot \sigma \cdot (T_2^4 - T_0^4) + h_0 (T_2 - T_0) \end{aligned}$$

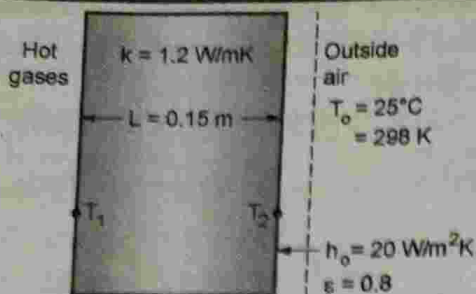


Fig. P. 1.21.15

$$\frac{1.2(T_1 - 373)}{0.15} = 0.8 \times (5.67 \times 10^{-8}) (373^4 - 298^4) + 20(373 - 298)$$

$$8(T_1 - 373) = 520.3 + 1500 = 2020.3$$

$$T_1 = 625.54 \text{ or } 352.54^\circ\text{C}$$

...Ans.

Ex. 1.21.16 : Derive an expression for temperature distribution and conduction heat flow in a conical rod with diameter at any section is given by, $D = A \cdot x$ where 'x' is the distance measured from the apex of the cone and A is certain numerical constant. Assume no heat generation and steady state and the lateral surface is well insulated.

Soln. : Refer Fig. P. 1.21.16

Fourier's law of heat conduction equation is,

$$Q = -k \cdot A_c \cdot \frac{dT}{dx}$$

$$= -k \times \frac{\pi}{4} D^2 \cdot \frac{dT}{dx}$$

$$Q = -k \cdot \frac{\pi}{4} (A^2 x^2) \frac{dT}{dx}$$

$$\therefore -k \cdot dT = \frac{4Q}{\pi \cdot A^2} \times \frac{dx}{x^2} \quad \dots(i)$$

Boundary conditions are :

$$\text{At } x = x_1, \quad T = T_1$$

and at

$$x = x_2, \quad T = T_2$$

On integrating Equation (i)

$$-k \int_{T_1}^{T_x} dT = \frac{4Q}{\pi A^2} \int_{x_1}^x \frac{dx}{x^2} = \frac{4Q}{\pi A^2} \left[-\frac{1}{x} \right]_{x_1}^x$$

$$-k(T_x - T_1) = \frac{4Q}{\pi A^2} \left[\frac{1}{x_1} - \frac{1}{x} \right]$$

$$\therefore T_x = T_1 - \frac{4Q}{\pi A^2 k} \left[\frac{1}{x_1} - \frac{1}{x} \right] \quad \dots(ii)$$

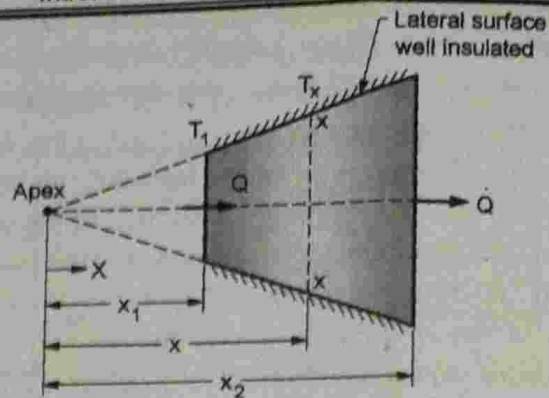


Fig. P. 1.21.16

For boundary conditions,

$$\text{At } x = x_1, \quad T = T_1 \text{ and at } x = x_2, \quad T = T_2$$

Again integrating the Equation (i) we get,

$$-k \int_{T_1}^{T_2} dT = \frac{4Q}{\pi \cdot A^2} \int_{x_1}^{x_2} \frac{dx}{x^2} = \frac{4Q}{\pi \cdot A^2} \left[-\frac{1}{x} \right]_{x_1}^{x_2}$$

$$-k(T_2 - T_1) = \frac{4Q}{\pi A^2} \left[\frac{1}{x_1} - \frac{1}{x_2} \right]$$

$$\therefore T_x = T_2 + \frac{4Q}{\pi A^2 k} \left[\frac{1}{x} - \frac{1}{x_2} \right] \quad \dots(iii)$$

On equating Equations (ii) and (iii)

$$T_1 - \frac{4Q}{\pi A^2 k} \left(\frac{1}{x_1} - \frac{1}{x} \right) = T_2 + \frac{4Q}{\pi A^2 k} \left(\frac{1}{x} - \frac{1}{x_2} \right)$$

$$\therefore T_2 = T_1 - \frac{4Q}{\pi A^2 k} \left[\frac{1}{x_1} - \frac{1}{x_2} \right]$$

$$\therefore \text{Heat flow rate, } Q = \frac{\pi A^2 k (T_1 - T_2)}{4 \left(\frac{1}{x_1} - \frac{1}{x_2} \right)} \quad \dots\text{Ans.}$$

On substituting the value of Q in Equation (ii), the temperature distribution is given as,

$$T_x = T_1 + (T_1 - T_2) \left[\frac{\left(\frac{1}{x} - \frac{1}{x_1} \right)}{\left(\frac{1}{x_1} - \frac{1}{x_2} \right)} \right] \quad \dots\text{Ans.}$$

Ex. 1.21.17 : A homogenous wall of area 'A' and thickness 'L' has left and right surface temperatures at 0°C and 40°C respectively. Determine the temperature at the centre of the wall.

- How much material must be added and to which side of wall if the temperature at the centre is raised by 5°C ?
- How much material must be removed and from which side of the wall if the temperature at the centre line of the wall is to be lowered by 5°C ?

Soln. : Refer Fig. P. 1.21.17.

Since, $Q = \frac{k A (T_2 - T_1)}{L}$... (i)

For temperature at the middle of the wall, we can write,

$$Q = \frac{k \cdot A (T_2 - T_c)}{\left(\frac{L}{2}\right)} \quad \dots (ii)$$

On equating Equations (i) and (ii)

$$\frac{k A (T_2 - T_1)}{L} = \frac{k \cdot A \cdot (T_2 - T_c)}{\left(\frac{L}{2}\right)}$$

$$T_2 - T_1 = 2 (T_2 - T_c)$$

$$\begin{aligned} \text{i.e. } T_c &= T_2 - \left(\frac{T_2 - T_1}{2}\right) = \frac{T_2 + T_1}{2} \\ &= \frac{40 + 0}{2} = 20^\circ\text{C} \quad \dots \text{Ans.} \end{aligned}$$

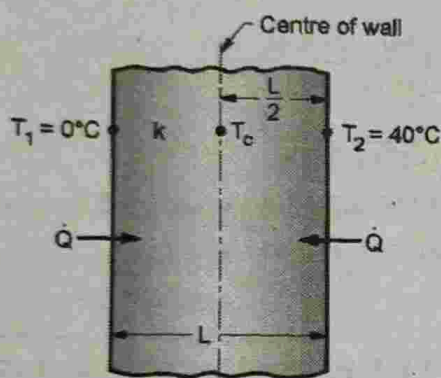


Fig. P. 1.21.17

(i) When temperature at the centre of wall is raised by 5°C

$$T_c = 20 + 5 = 25^\circ\text{C}$$

Let x be the thickness of material added right side of the wall. Then modifying the Equation (i) we get,

$$Q = \frac{k A (T_2 - T_1)}{L + x} \quad \dots (iii)$$

For middle temperature

$$Q = \frac{k A (T_2 - T_c)}{\frac{L}{2} + x} \quad \dots (iv)$$

On equating Equations (iii) and (iv)

$$\frac{k A (T_2 - T_1)}{L + x} = \frac{k A (T_2 - T_c)}{\frac{L}{2} + x}$$

$$\frac{(40 - 0)}{L + x} = \frac{(40 - 25)}{\frac{L}{2} + x} \quad \dots (v)$$

$$\text{On solving, } x = -\frac{L}{5}$$

Negative sign indicates that the assumption of adding the wall on the right side is incorrect.

Thus the material of thickness $x = \frac{L}{5}$ must be added towards the left side of the wall so that $T_c = 25^\circ\text{C}$... Ans.

(ii) When T_c is lowered by 5°C

$$\text{i.e. } T_c = 20 - 5 = 15^\circ\text{C}$$

Let x be the material removed from right side. Again modifying the Equations (iii) and (iv) by equating,

$$\frac{k A (T_2 - T_1)}{L - x} = \frac{k A (T_2 - T_c)}{\frac{L}{2} - x}$$

$$\frac{40 - 0}{L - x} = \frac{40 - 15}{\frac{L}{2} - x}$$

$$20L - 40x = 25L - 25x$$

$$15x = -5L$$

$$x = -\frac{5L}{15} = -\frac{L}{3}$$

Negative sign shows that the material of $\frac{L}{3}$ thickness be removed from left side instead of from right side to get 15°C temperature at the middle of the wall. ... Ans.

Ex. 1.21.18 : A 60 W lamp is buried in soil ($k = 0.83 \text{ W/mK}$) at 0°C and switched on. Find the temperature 0.3m away from the lamp, when steady state is reached.

Soln. :

$Q = 60 \text{ W}$. The heat is dissipated by lamp considering it as spherical in shape through the soil. Let T_1 be the temperature at 0.3 m radius from Fourier's law,

$$Q = -k \cdot A \cdot \frac{dT}{dr} = -k \cdot (4\pi r^2) \cdot \frac{dT}{dr}$$

$$Q \cdot \frac{dr}{r^2} = -k \cdot 4\pi \cdot dT \quad \dots (i)$$

Boundary conditions are ;

$$\text{At } r = 0.3; T = T_1$$

$$\text{At } r = \infty; T = 0$$

On integrating equation (i) within boundary conditions,

$$\int_{0.3}^{\infty} \left(\frac{dr}{r^2}\right) = -k \cdot 4\pi \int_{T_1}^0 dT$$



$$Q \left[-\frac{1}{r} \right]_{0.3}^{\infty} = -4\pi k \cdot [T]_{T_1}^0$$

$$60 \left[-\frac{1}{\infty} + \frac{1}{0.3} \right] = -4\pi \times 0.83 (0 - T_1)$$

$$T_1 = 19.175^\circ\text{C} \quad \dots \text{Ans.}$$

Ex. 1.21.19 : Heat is conducted through a tapered circular rod of 200 mm length. The ends A and B having diameters 50 mm and 25 mm are maintained at 27°C and 227°C respectively. k (rod material) = $40 \text{ W/m}^\circ\text{C}$. Find :

- Heat conducted through the rod.
 - The temperature at the mid-point of the rod.
- Assume one dimensional steady state conduction.

Soln. :

Given : $T_A = 27^\circ\text{C}$, $T_B = 227^\circ\text{C}$, $k = 40 \text{ W/m}^\circ\text{C}$

$D_A = 50 \text{ mm}$; $D_B = 25 \text{ mm}$;

Consider section of rod y-y at distance x from A end and diameter be called at D_x at y-y section.

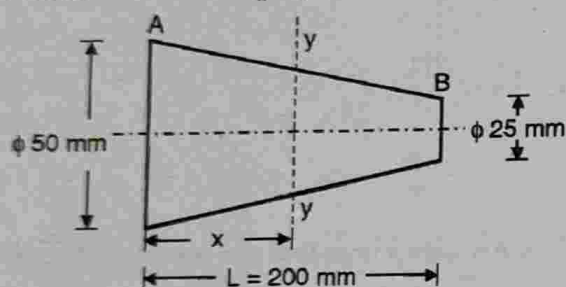


Fig. P. 1.21.19

$$\therefore D_x = D_A - \left(\frac{D_A - D_B}{L} \right) x$$

$$= 0.05 - \left(\frac{0.05 - 0.025}{0.2} \right) x$$

$$D_x = 0.05 - 0.125 x$$

$$\therefore A = \frac{\pi}{4} D_x^2 = \frac{\pi}{4} (0.05 - 0.125 x)^2$$

Now, $Q = -kA \frac{dT}{dx}$ considering only x-directional heat flow (given in question)

$$= -40 \times \frac{\pi}{4} (0.05 - 0.125 x)^2 \cdot \frac{dT}{dx}$$

$$\frac{4Q}{40\pi} \times \frac{dx}{(0.05 - 0.125 x)^2} = -dT$$

Boundary conditions

(1) At $x = 0$ $T = T_A = 27^\circ\text{C}$

(2) At $x = 0.2$ $T = T_B = 227^\circ\text{C}$

Integrating

$$\frac{4Q}{40\pi} \int_0^{0.2} \frac{dx}{(0.05 - 0.125 x)^2} = - \int_{27}^{227} dT$$

$$\frac{4Q}{40\pi} \left[-\frac{1}{(0.05 - 0.125 x)(-0.125)} \right]_0^{0.2} = -[T]_{27}^{227}$$

$$\frac{4Q}{0.125 \times 40\pi} \left[\frac{1}{0.05 - 0.125 \times 0.2} - \frac{1}{0.05} \right] = -[227 - 27]$$

$$Q \times 0.255 \left[\frac{1}{0.025} - \frac{1}{0.05} \right] = -200$$

$$Q \times 0.255 [40 - 20] = -200$$

$$Q = -39.22 \text{ W} \quad \dots \text{Ans.}$$

Negative sign indicates that heat is flowing from B to A end.

Temperature of midpoint $x = 0.1 \text{ m}$

$$\frac{4Q}{40\pi} \int_0^{0.1} \frac{dx}{(0.05 - 0.125 x)^2} = - \int_{27}^{T_x} dT$$

$$\frac{4Q}{0.125 \times 40\pi} \left[\frac{1}{0.05 - 0.125 \times 0.1} - \frac{1}{0.05} \right] = -(T_x - 27)$$

$$Q \times 0.255 \left[\frac{1}{0.0375} - \frac{1}{0.05} \right] = -(T_x - 27)$$

$$Q \times 0.255 [26.667 - 20] = -(T_x - 27)$$

$$-39.22 \times 0.255 \times 6.667 = -T_x + 27$$

$$T_x = 93.68^\circ\text{C} \quad \dots \text{Ans.}$$

Ex. 1.21.20 : A surface having an area of 1.5 m^2 and maintained at 300°C exchanges heat by radiation with another surface at 40°C . The value of factor due to the geometric location and emissivity is 0.52. Determine :

- Heat lost by radiation
- The value of thermal resistance and
- The value of equivalent convection coefficient.

Soln. :

Given : $A_1 = 1.5 \text{ m}^2$; $T_1 = 300^\circ\text{C} = 573 \text{ K}$

$T_2 = 40^\circ\text{C} = 313 \text{ K}$; $F_{12} = 0.52$

1. Heat flow rate by radiation,

$$Q_{12} = F_{12} \sigma (T_1^4 - T_2^4) A_1$$

$$= 0.52 \times 5.67 \times 10^{-8} (573^4 - 313^4) \times 1.5$$

$$= 4343 \text{ W} \quad \dots \text{Ans.}$$

2. For thermal resistance,

$$Q = \frac{\Delta T}{R} = F_{12} \sigma (T_1 - T_2) (T_1 + T_2) (T_1^2 + T_2^2) \times A$$

$$= \frac{T_1 - T_2}{\frac{1}{F_{12} \sigma (T_1 + T_2) (T_1^2 + T_2^2) A_1}} = \frac{\Delta T}{R}$$

$$\therefore R = \frac{1}{0.52 \times 5.67 \times 10^{-8} (573 + 313) (573^2 + 313^2) \times 1.5}$$

$$\therefore R = 0.06 \text{ K/W} \quad \dots \text{Ans.}$$

3. Equivalent convection coefficient

$$Q = hA\Delta T$$

$$h = \frac{4343}{1.5 \times (573 - 313)} = 11.14 \text{ W/m}^2 \text{ K} \dots \text{Ans.}$$

Ex. 1.21.21 : Calculate the rate of heat loss for a red brick wall of length 5 m, height 4 m and thickness 25 cm. Temperature of the inner surface is 110°C and that of the outer surface is 40°C . Thermal conductivity of the red brick is 0.7 W/mK . Also, calculate the temperature at an interior point of the wall (in the thickness), 20 cm away from the inner surface of the wall.

Soln. :

$$\text{Given : } T_1 = 110^\circ\text{C}; \quad T_2 = 40^\circ\text{C};$$

$$\Delta x = 25 \text{ cm} = 0.25 \text{ m}; \quad A = 5 \text{ m} \times 4 \text{ m}; \quad k = 0.7 \text{ W/mK}$$

Heat flow rate across the wall :

$$Q = \frac{T_1 - T_2}{\frac{\Delta x}{kA}} = \frac{110 - 40}{\frac{0.25}{0.7 \times 5 \times 4}} = 3920 \text{ W} \quad \dots \text{Ans.}$$

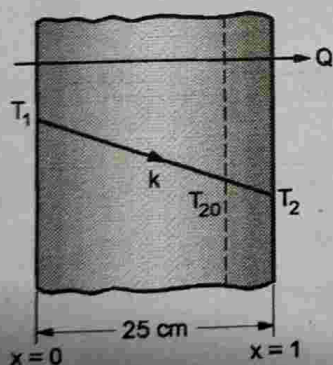


Fig. P. 1.21.21

To find temperature in thickness

We know that temperature profile can be found out from $\frac{d^2T}{dx^2} = 0$ for one dimensional heat flow in x-direction.

$$\text{Integrating } \frac{dT}{dx} = C_1 \quad \dots (i)$$

$$\text{and } T = C_1x + C_2 \quad \dots (ii)$$

applying boundary conditions,

$$T_1 = C_1 \times 0 + C_2 \Rightarrow C_2 = 110$$

$$T_2 = C_1 \times 0.25 + 110 = 40$$

$$\therefore C_1 = -280$$

Substituting C_1 and C_2 and $x = 20 \text{ cm}$ in Equation (ii)

$$T_{20} = -280 \times 0.2 + 110 = 54^\circ\text{C} \quad \dots \text{Ans.}$$

Ex. 1.21.22 : A 10 cm diameter copper ball is to be heated from 100°C to an average temperature of 150°C in 30 minutes. Taking average density and specific heat of copper in this temperature range as 8950 kg/m^3 and $0.395 \text{ kJ/kg}^\circ\text{C}$ respectively, determine :

- The total amount of heat transfer to the ball and
- The average rate of heat transfer to the ball and average heat flux.

Soln. :

$$D = 10 \text{ cm} = 0.1 \text{ m}$$

$$\rho = 8950 \text{ kg/m}^3, \quad C_p = 0.395 \text{ kJ/kg}^\circ\text{C}$$

- Total heat transfer to ball**

$$Q = m C_p \Delta T = \rho V C_p \Delta T$$

$$= 8950 \times \frac{4}{3} \pi \times (0.05)^3 \times 395 \times (150 - 100)$$

$$Q = 92505.7 \text{ J (In 30 minutes)} \quad \dots \text{Ans.}$$

- Average heat transfer rate to ball**

$$= \frac{92505.7}{30 \times 60} = 51.39 \text{ W} \quad \dots \text{Ans.}$$

Average heat flux

$$q = \frac{51.39}{4\pi \times 0.05^2} = 1.637 \text{ W/m}^2 \quad \dots \text{Ans.}$$

Ex. 1.21.23 : The roof of an electrically heated room is 6 m long, 8 m wide and 0.25 m thick, and is made of a flat layer of concrete whose thermal conductivity is $0.8 \text{ W/m}^\circ\text{C}$. The temperatures of inner and outer surfaces of room one night are measured to be 15°C and 4°C respectively, for a period of 10 hours. Determine -

- Rate of heat loss through roof that night and
- Cost of heat loss if cost of electricity is Rs. 4/kWh

Soln. :

$$\text{Area } A = 6 \text{ m} \times 8 \text{ m} = 48 \text{ m}^2;$$

$$\Delta x = 0.25 \text{ m};$$

$$K = 0.8 \text{ W/m}^\circ\text{C},$$

$$\text{Period} = 10 \text{ hrs.}$$

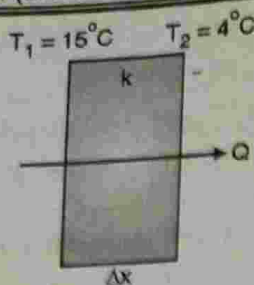


Fig. P. 1.21.23

(i) Rate of heat loss

$$Q = \frac{T_1 - T_2}{\frac{\Delta x}{kA}} = \frac{15 - 4}{\frac{0.25}{0.8 \times 48}}$$

$$Q = 1689.6 \text{ W} \quad \dots \text{Ans.}$$

(ii) Heat loss for 10 hours

$$Q = \frac{1689.6}{1000} (\text{kW}) \times 10 (\text{h})$$

$$= 16.896 \text{ kWh}$$

$$\text{Cost of heat loss} = 16.896 \times 4 = \text{Rs. } 67.58 \quad \dots \text{Ans.}$$

Ex. 1.21.24 : A uniformly tapered rod of square cross-section is shown in Fig. P. 1.21.24. Heat is conducted along its X-axis at uniform rate. The thermal conductivity of the material is 40 W/mK.

Find :

- The rate of heat conduction.
- Temperature at 40 cm distance from left face.

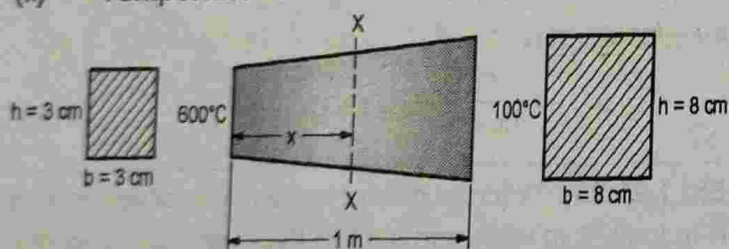


Fig. P. 1.21.24

Soln. : $k = 40 \text{ W/mK}$;

Let h_x be the height of the section at a distance x from left face.

$$h_x = 0.03 + \left(\frac{0.08 - 0.03}{1} \right) \cdot x = 0.03 + 0.05x$$

$$\text{Surface area, } A_x = h_x \times b_x = h_x^2 \quad (\because \text{section is square})$$

$$= (0.03 + 0.05x)^2$$

(i) Heat flow rate, Q

From Fourier's law of heat conduction,

$$Q = -kA \cdot \frac{dT}{dx}$$

$$\frac{Q}{A} \cdot dx = -k \cdot dT$$

$$\frac{Q}{(0.03 + 0.05x)^2} \cdot dx = -40 \cdot dT \quad \dots (i)$$

The boundary conditions are,

$$\text{At } x = 0, \quad T = 600^\circ\text{C}$$

$$\text{At } x = 1 \text{ m, } T = 100^\circ\text{C}$$

On integrating both sides,

$$Q \int_0^1 \frac{dx}{(0.03 + 0.05x)^2} = -400 \int_{600}^{100} dT \quad \dots (ii)$$

$$Q \left[\frac{-1}{(0.03 + 0.05x)} \right]_0^1 = -400 [T]_{600}^{100}$$

$$-Q \left[+\frac{1}{0.03 + 0.05} - \frac{1}{0.03} \right] = -400 (100 - 600)$$

$$-Q \left[\frac{1}{0.08} - \frac{1}{0.03} \right] = +400 \times 500$$

$$20.833Q = 400 \times 500$$

$$Q = 9600 \text{ W} \quad \dots \text{Ans.}$$

(ii) Temperature T at $x = 40 \text{ cm} = 0.4 \text{ m}$

At $x = 0$, $T = T_1 = 600^\circ\text{C}$ and at $x = 0.4 \text{ m}$, $T = T_x$ on changing the boundary conditions in Equation (ii) we get,

$$Q \int_0^{0.4} \frac{dx}{(0.03 + 0.05x)^2} = -400 \int_{600}^{T_x} dT$$

$$Q \left[-\frac{1}{(0.03 + 0.05x)} \right]_0^{0.4} = -400 (T_x - 600)$$

$$-9600 \left[\frac{1}{0.03 + 0.05 \times 0.4} - \frac{1}{0.03 + 0.05 \times 0} \right]$$

$$= 400 [600 - T_x]$$

$$-9600 \left[\frac{1}{0.05} - \frac{1}{0.03} \right] = 400 [600 - T_x]$$

$$128000 = 400 (600 - T_x)$$

$$320 = 600 - T_x$$

$$T_x = 280^\circ\text{C} \quad \dots \text{Ans.}$$

Ex. 1.21.25 : A truncated cone which is 30 cm high is made of aluminium with $k = 220 \text{ W/mK}$. The diameter of the top surface = 7.5 cm. Diameter of bottom surface = 12.5 cm. The lower surface is maintained at 93°C , while the upper surface is maintained at 540°C . The curved lateral surface is insulated. Assuming one dimensional axial heat flow, what is the rate of heat flow at steady state ?

Soln. : Refer Fig. P. 1.21.25.

$$T_1 = 540^\circ\text{C}; \quad T_2 = 93^\circ\text{C}$$

$$k = 220 \text{ W/mK}; \quad d_1 = 7.5 \text{ cm}; \quad d_2 = 12.5 \text{ cm}$$

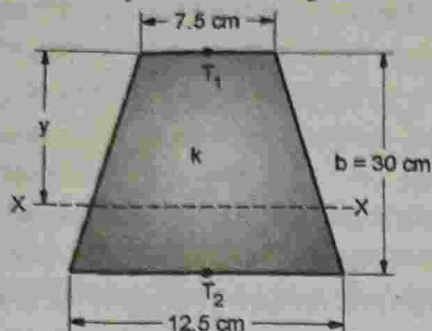


Fig. P. 1.21.25

Consider any section (X-X) at distance y and let the diameter of the cone be D_y .

$$\begin{aligned} D_y &= d_1 + \frac{(d_2 - d_1)}{b} \times y \\ &= 0.075 + \left(\frac{0.125 - 0.075}{0.30} \right) y \\ &= 0.075 + 0.166 y \end{aligned}$$

$$A = \frac{\pi}{4} D_y^2 = \frac{\pi}{4} (0.075 + 0.166 y)^2$$

From Fourier's equation,

$$\begin{aligned} Q &= -kA \cdot \frac{dT}{dy} \\ &= -220 \times \frac{\pi}{4} \cdot (0.075 + 0.166 y)^2 \cdot \frac{dT}{dy} \end{aligned}$$

$$\therefore \frac{4Q}{220\pi} \cdot \frac{dy}{(0.075 + 0.166 y)^2} = -dT$$

On integrating the above equation between the boundary conditions as,

$$\text{At } y = 0, \quad T = T_1 = 540^\circ\text{C} \text{ and}$$

$$\text{At } y = 0.3 \text{ m}, \quad T = T_2 = 93^\circ\text{C}$$

$$\therefore \frac{4Q}{\pi \times 220} \int_0^{0.3} \frac{dy}{(0.075 + 0.166 y)^2} = - \int_{540}^{93} dT$$

$$\frac{4Q}{\pi \times 220} \left[-\frac{1}{0.166 (0.075 + 0.166 y)} \right]_0^{0.3} = -[T]_{540}^{93}$$

$$-\frac{4Q}{0.166 \times \pi \times 220} \left[+\frac{1}{0.075 + 0.166 \times 0.3} - \frac{1}{0.075} \right] = -(93 - 540)$$

$$-\frac{4Q}{\pi \times 220} \left[\frac{1}{0.1248} - \frac{1}{0.075} \right] = 447$$

$$Q = 2409.8 \text{ W}$$

...Ans.

Ex. 1.21.26 : A gas filled tube has 2 mm inside diameter and 25 cm length. The gas is heated by an electrical wire a diameter 0.05 mm located along the axis of tube. Current and voltage drop across the heating element are 0.5 amperes and 4 volts respectively. If the wire temperature is 175°C and inside tube wall temperature is 150°C respectively, find the thermal conductivity of the gas filling the tube.

Soln. :

$$\text{Heat flow rate } Q = V \times I = 4 \times 0.5 = 2 \text{ W}$$

$$D = 2 \text{ mm}, \quad r_2 = 1 \text{ mm} = 0.001 \text{ m}$$

$$d = 0.05 \text{ mm}; \quad r_1 = 0.000025 \text{ m}$$

$$Q = \frac{\Delta T}{R} = \frac{175 - 150}{\frac{\ln \frac{r_2}{r_1}}{2\pi k L}} = 2$$

$$\text{or } \frac{25 \times 2\pi k \times 0.25}{\ln \left(\frac{0.001}{0.000025} \right)} = 2 \Rightarrow k = 0.188 \text{ W/mK} \quad \dots \text{Ans.}$$

Summary

- The science of thermodynamics deals with the energy transfers and its effect on the condition of the system while the science of heat transfer deals with the determination of rates of transfer of heat energy across the system.
- Knowledge of heat transfer helps us to determine (i) the rates of heat transfer of a given temperature difference, (ii) time taken to affect the desired amount of heat energy, (iii) surface area needed and (iv) temperature distribution across the system.
- Heat transfer plays an important role in various industrial applications. e.g. design of heat exchangers and ducts in air conditioning; cylinder and radiators in I.C. engines; combustion chambers; steam generators; condensers; transformers; furnaces; various electronic system etc.
- Modes of heat transfer are conduction, convection and radiation.
- Conduction of heat is defined as the transfer of heat from one part of substance to another part of substance by direct contact of molecules.
- The process of heat convection is due to the capacity of moving matter to carry heat energy.

Exercise

[Note : For Answers please refer the section number indicated in bracket.]

Theory

- Q. 1** Define the terms heat, thermodynamics and heat transfer. [Section 1.1]
- Q. 2** Name and explain the various modes of heat transfer. [Section 1.2]
- Q. 3** Differentiate between conductive and convective heat transfers. [Section 1.2]
- Q. 4** Write a short note on thermal conductivity of gases. [Section 1.5.4 (5)]
- Q. 5** How does thermal conductivity of gases and liquids vary with temperature.
[Section 1.5.4 (5) and (6)]
- Q. 6** Comment upon variations of thermal conductivity of thermally conducting and insulating materials with temperature.
[Sections 1.5.4 (3) and 1.5.4 (4)]
- Q. 7** Explain in brief the analogy between the heat flow and electricity with its significance.
[Section 1.19]
- Q. 8** Discuss the mechanism of thermal conduction in gases and solids. Name some good conductors and poor conductors of heat.
[Section 1.3 and 1.5.4 (3)]
- Q. 9** Write short notes on :
(i) Concept of black body. [Section 1.8]
(ii) Newton's law of cooling by convection.
[Section 1.7.2]
"Stefan-Boltzmann law"
- Q. 10** Write short note on "Stefan-Boltzmann law"
[Section 1.9.3]
- Q. 11** What are boundary condition for conduction?
[Section 1.6]
- Q. 12** Discuss Fourier's law of heat conduction, why negative sign inserted in its expression.
[Section 1.6]
- Q. 13** State Kirchoff's law. [Section 1.9.2]
- Q. 14** Derive the general heat conduction equation in Cartesian co-ordinates for isotropic materials.
[Section 1.10]
- Q. 15** Prove that the heat conduction through a hollow long cylinder is given by :
$$Q = \frac{T_1 - T_0}{\ln \left(\frac{r_2}{r_1} \right)} \frac{2\pi k L}{\text{[Section 1.19]}}$$
- Q. 16** Derive the equation for thermal resistance for heat conduction in hollow sphere.
[Section 1.20]